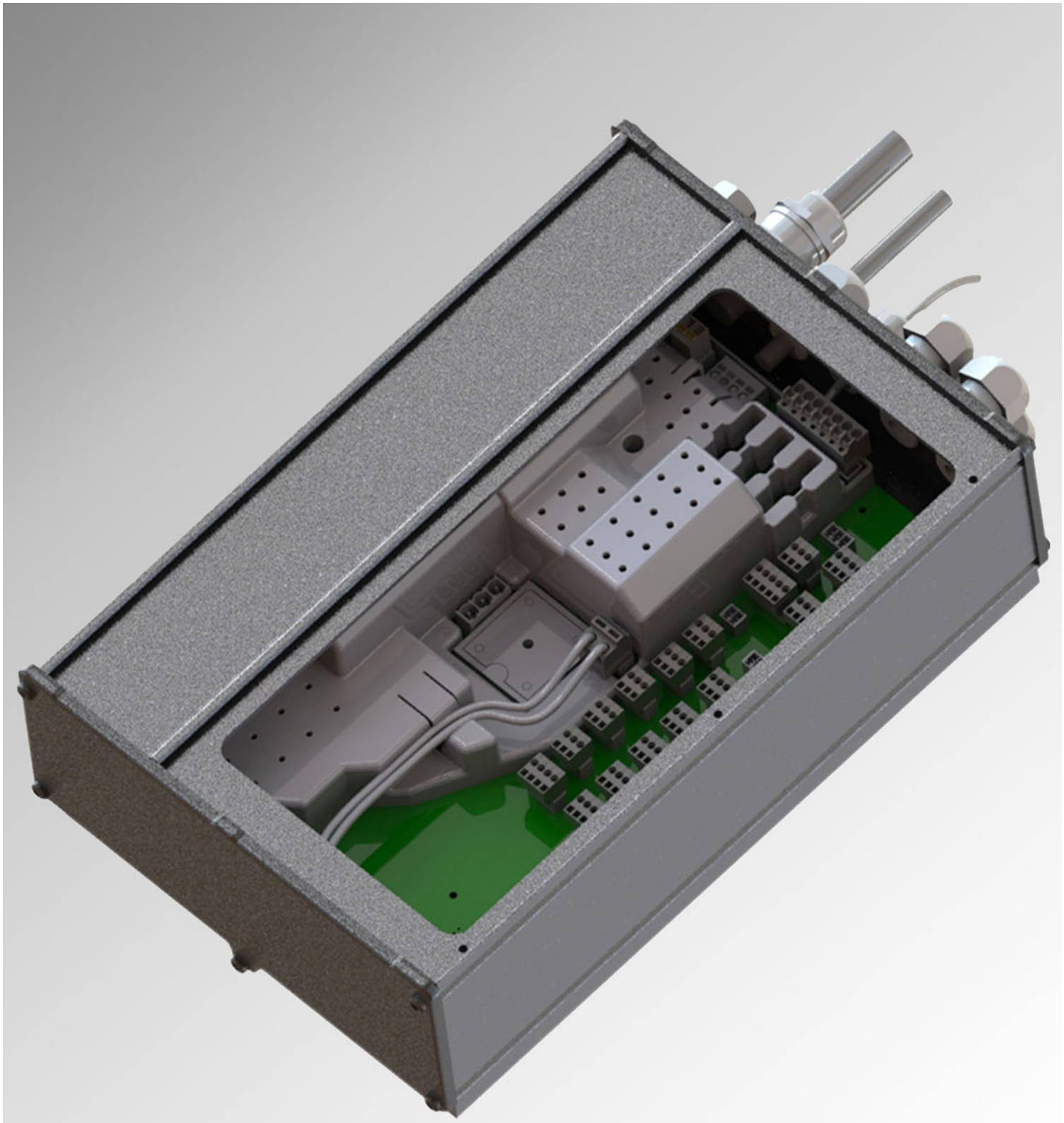
User Manual
Control system
ASSA ABLOY MCC 2 (Type 7110)

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Our relationship continues

As your entrances are part of your business flow, there's every reason to keep them working well. ASSA ABLOY Entrance Systems offers you a maintenance and modernization expertise to rely on. Our Maintenance Programs and Modernization Services for entrance automation is backed by a extensive expertise for all types of pedestrian- and industrial door and docking systems, independent of brand. At your disposal is a team of dedicated expert technicians, proven through decades of maintenance, service and satisfied customers.

Maintenance of your door

To make sure your ASSA ABLOY door continues to operate with maximum reliability, safety and performance, it is necessary to do regular maintenance as specified in this user manual and in the logbook supplied with your ASSA ABLOY door. This maintenance must be done by qualified service engineers trained to recognise possible problems, and to replace parts that have a defined life expectancy.

Dismantling and disposal of the door

Dismantling of the door or parts thereof may only be carried out by qualified personnel. These tasks are complicated and potentially dangerous. Please make a contribution to protecting the environment when disposing of the door at the end of its functional life. Contact your local ASSA ABLOY Entrance Systems Service Centre for appropriate means of disposal.

Your local service centre

It is recommended that you contact your local ASSA ABLOY Entrance Systems Service Centre for details about ASSA ABLOY Entrance Systems Pro-Active care progams available for your ASSA ABLOY high speed door.

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1. Introduction

These operating instructions apply to the use of the frequency converter control system MCC 2 (Type 7110) Vector Control with high speed doors. For the sake of simplicity, the control system is called only MCC 2 throughout this document. They are part of the product and describe the intended and safe use throughout the whole product life cycle.
Before the installation of the MCC 2 read the installation and operating instructions carefully and familiarize yourself with their contents.
The manufacturer is not liable for damage caused by not observing of the operating instructions as well as their safety instructions, or a not intended use of the MCC 2.

1.1 Applicable documents

These operating instructions are only valid in combination with the operating instructions for the ASSA ABLOY high-speed door and the inspection book. Both of these documents are supplied with the product.

1.2 Declaration of conformity

The control system is an integral part of the door. The declaration of conformity applies to the high-speed door as well as the control unit. It is included in the inspection book supplied with the door.
The control system is designed in compliance with the requirements of the EN 13241-1 standard for door products and the subordinate standards mentioned in this standard.
Modifications that affect the technical data specified in the operating instructions or intended use, i.e. modifications that materially alter the machine, invalidate the manufacturer's declaration of conformity.

1.3 Amendments and validity

The information contained in these instructions constitutes the technical specifications applicable at time of publication. Significant changes will be incorporated in a new edition of the operating instructions. The document and version number of these instructions can be found in the cover page.

1.4 Document storage

Keep these operating instructions in the storage bag inside the control cabinet or in the supplied document box.

1.5 Target group

These operating instructions are addressed to trained, qualified and authorised personnel. All setup configurations, and in particular repairs and maintenance, may only be performed by employees of the manufacturer or other trained specialists.

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1.6 Signal words



Danger
Indicates a danger which, if not avoided, will result in serious injury or death.



Warning
Indicates a electrical danger which, if not avoided, may result in serious injury or death.



Caution
Indicates a danger which, if not avoided, may result in minor or moderate injury.



Notice
Indicates important information (e.g. material damage), but does not indicate dangers.

2. Safety instructions

Observe the safety instructions in chapter 2 and the safety instructions in the user manual for the door.

2.1 Intended use

The control is exclusively intended for opening and closing a power-operated high-speed door with a drive.
Never make any modifications or changes to the product without the express written authorization of the manufacturer.

2.2 Foreseeable misuse

Any use other than described in chapter Intended use is regarded as reasonably foreseeable misuse. This includes but is not limited to:

- the use on overhead or sliding doors.

Any damage or injury as a result of reasonably foreseeable misuse or of not following the assembly and operating instructions will render the manufacturer's liability null and void.

2.3 Personnel qualifications

The following persons are qualified to perform assembly work and to work on the mechanical system (troubleshooting & repair):

- skilled workers with relevant training, e.g. industrial mechanic

A skilled worker is a person who, due to his/her professional training, his knowledge and experience as well as due to his/her knowledge of the relevant regulations, is able to judge the work assigned to him/her as well as to identify possible hazards.
The following persons are qualified to perform electrical installation work and to work on the electrical system (troubleshooting, repair & deinstallation):

- qualified electricians

Skilled electricians must be able to read and understand electric circuit diagrams, to put electrical systems into service and to maintain them, to wire control cabinets, to install the control software, to ensure the functionality of electrical components and to identify possible hazards resulting from handling electrical and electronic systems.
The following persons are authorized to handle the product:

- operating personnel

The operator must have read and understood the instructions, in particular the "Safety" chapter and must be aware of the hazards associated with handling the product and the controlled door system. The operator must have been instructed with regard to handling the controlled door system.

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2.4 General safety instructions

- Danger may be posed by this product and by the controlled door system when it is used improperly, incorrectly, or for purposes other than the intended use.
- Keep all safety information and instructions for future reference.
- Only use genuine spare parts of the manufacturer. Wrong or faulty spare parts can cause damage, malfunctions or even a total failure of the product.
- Observe the general, legal and other national valid regulations for accident prevention and acknowledged technical rules for safety-oriented work.
- Only operate the door, if it is equipped with a disconnect device for the main supply voltage (such as a suitable main switch or a CEE plug). The main power disconnect device must be easily accessible and located between 0.6 and 1.7 meters above the access level.
- Make no change to the control or the safety devices which may affect the safety of the door.
- Some parameters are protected by an access code for safety reasons. The necessary access code level is given in square brackets after the chapter heading. Detailed information about the access code can be found in chapter Access codes MCC 2 - Display.
- Only operate the door in hold to run mode, if the separate emergency stop device is mounted and installed next to the display unit.

2.5 Safety instructions for operating the door

Before operating the door make sure that:

- The door is assembled in accordance with the installation instructions.
- All wiring between the door, the control system and the drive is connected properly.
- All safety devices and protection equipment are installed as intended use and are functional.
- Safety devices are not modified or out of operation.

Leave the main power on at all times, to avoid the build-up of condensation in the control system housing.

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2.6 Safety instructions for installation and maintenance



Danger: Hazardous voltage!

Fatal electric shock when touching live parts. Observe the following safety rules when working on the electrical system.
Disconnect from the mains.
Secure against inadvertent switch-on.
Verify de-energized state.
Work on the electrical system may only be performed by skilled electricians or instructed persons working under the direction and supervision of a skilled electrician in accordance with the electrotechnical rules and directives

- The electrical work must be carried out according to the applicable local standards and legal regulations.
- Before performing any work on the control system, cables, connectors or electrically live components, always switch off and lock the main switch or disconnect and secure the CEE plug.
- Ensured that neither people nor vehicles pass through the door or are in the door hazard area during installation and maintenance.
- Dangerous touch voltage applies at the control up to 30 seconds after switching off/ disconnecting the supply voltage (discharge of the capacitors). Therefore wait min. 30 seconds before working on electrical parts, wiring or opening the MCC 2. The non-observance can lead to an electric shock with serious injuries or death.
- Generally, avoid touching electronic components on the MCC 2 Board.
- If it is necessary to work on electronic components use appropriate ESD protective equipment before touching the circuit board of the control system and the display, in order to avoid damages by electronic discharge,
- When opening the housing cover of the MCC 2, protect the control system against moisture.

2.7 Safety instructions for "hold to run mode"



Warning: Crush hazard and risk of being struck by the closing door!

Persons can be struck when the door is closed or collide with the door.
The door must be visible from the place of operation.

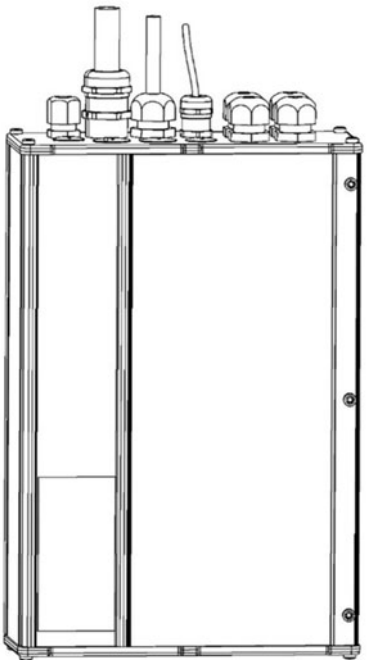
- In hold to run mode all safety devices are disabled. The door moves as long as the opening or closing button is pressed and the door has not reached the upper or lower end position. The door does not switch off until it reaches the upper or lower end position.
- An additional emergency stop device needs to be mounted and installed next to the display unit. If the display unit fails in the hold to run mode, the door can be stopped safely at any time using the additional emergency stop switch (details see chapter 3.1.2.4).
- Do not operate the door if any person or object is located in the door hazard area.
- The door must be visible from the place of operation.

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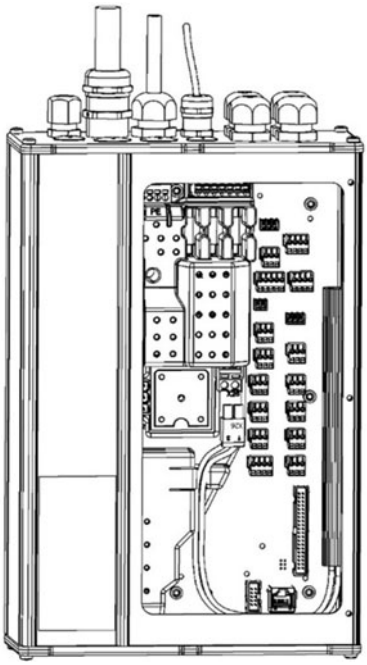
3. Product Description

The MCC 2 control system is designed to be used with high speed doors. The MCC 2 meets the valid and applicable European CE requirements, which has been tested and confirmed by independent testing institutes. The control system contains all necessary components to operate these doors with consideration of all safety criteria.
The characteristics of the MCC 2 Control System with Vector Control allow even smoother starting and stopping of doors than is possible with standard frequency converters. This protects the door's mechanism against hard impacts and increases its speed.

3.1 MCC 2 main unit with housing



Control system MCC 2 closed
Figure 1



Control system MCC 2 opened
Figure 2

The door control is built in an anodized aluminium housing. There is a connecting plate at each end for the introduction of the cables and for the junction box to be mounted.
Inside the housing (see Figure 2) the following components are located on the control board:

- The plug-in connection terminals for the inputs and outputs of the control system.
- The connection terminals for the net or primary supply power, motor, brake and PFC filter cables.
- 3 integrated pre-fuses (10 A fast Schurter SHF)
- The frequency converter including heat sink
- The short-circuit-proof power supply for the supply of the complete microprocessors, the frequency converter and external 24 V loads.
- Brake control

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3.1.1 Safety devices of the control system

The closing movement of an automatic door is monitored by various safety devices in order to avoid accidents. The safety devices and functions incorporated in the control system are described below.
The doors are also equipped with other safety devices, depending on the door model. See the 'Safety Systems' section of the door user manual for information on the safety devices provided with the various models.
As an electric safety device, a pre-running safety photocell, light curtain or an electrical safety edge can be used. The pre-running safety photocell and the electrical safety edge are complemented by a door line photocell in the side frame.
The electrical safety circuits of the MCC 2 control system fulfil the requirements of the product standard for doors EN 13241-1.

3.1.1.1 Testing the safety devices

Before each closing movement of the door the fault free operation of the safety devices and the door line photocell are automatically tested. If a fault is detected, no release of the closing movement occurs and an error message is displayed by the control. In case of an error the control changes in the hold to run mode, which allows the manual operation of the door.
Electrical safety edges are tested continuously.
Doors with light curtains are tested once per cycle.



Observe the safety instructions for "hold to run mode" (chapter Safety instructions for "hold to run mode").

3.1.1.2 Electrical safety edge (only for some types of door)

The electrical safety edge serves as closing edge protection. It consists of a rubber profile fitted to the main closing edge of the door. Any deformation of the rubber profile initiates an electrical impulse. This impulse immediately stops the door while it is closing and reverses regardless of the set opening height, up to the reference limit switch and is locked there. Locking is indicated by the following message on the display.



Pressing the "Open" or "Close" buttons releases the locking of the door.
However, you can adjust the function such, that the door automatically closes after a set time after the electrical safety edge has been activated (chapter Adjustment of activations of the safety edge or pre-running photocell [2]).
Just before the lower end position of the door (max. 50 mm – 2") the electrical safety edge is switched off because otherwise an activation via the ground would take place.
For classifications of safety circuits to EN ISO 13849-1 see chapter Classifications of safety circuits to EN ISO 13849-1.

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3.1.1.3 Pre-running safety photocell (only for some door types)

The pre-running safety photocell runs approx. 100 mm (4") below the bottom profile (depending on the door type). It moves with the door as it opens and closes. It retracts into the bottom profile when the door is in the lower end position. At the lower end position of the door the pre-running safety photocell is switched off because otherwise an activation via the ground would take place. The switching off point depends from the door type.
If the pre-running safety photocell is activated while the door is closing the door stops immediately and then opens again until it reaches the reference limit switch (reversing). After the tenth activation of the pre-running safety photocell (without the door closing completely) the door will be locked in the upper end position. However, you can parameterize the function differently (chapter Adjustment of activations of the safety edge or pre-running photocell [2]).
For classifications of safety circuits to EN ISO 13849-1 see chapter Classifications of safety circuits to EN ISO 13849-1.

3.1.1.4 Door line photocell

The door line photocell is mounted in the lower area of the side frame of the door where it is protected. The proper function is monitored also by the control system. An activation of the photocell while the door is open resets and starts the hold-open time or the close after passage time. An activation of the photocell while the door is closing the door stops and then opens again until it reaches the reference limit switch (reversing). Unlike for the electrical safety edge, the door is not locked in this end position, instead it closes again automatically after the hold-open or close after passage time, if the door was closing in auto close mode while the photocell was activated. Otherwise it will stay open. As an option it's possible to get at some door types a light curtain instead of the door line photocell.

3.1.1.5 Emergency stop

The MCC 2 control system has provision for connecting two separate 2-channel emergency stop devices corresponding to Stop Category 0 (stopping by immediate interruption of the power supply to the motor). The MCC 2 input for this fulfils the requirements of Category 3 (2-channel, Performance Level d of the EN ISO 13849-1). To reach this level, the wiring an application notes in chapter Emergency stop input 1 (X72) and OSSD Emergency stop input 2 (X73) must be observed!

3.1.2 Operating modes

The control system has several operating modes, which you can select by means of the menus. The following are the main operation modes.

3.1.2.1 Open by impulse / close after time

After an Open impulse, the door travels until it reaches the door open end position. The closing sequence starts after the hold-open time expires. The door closes until it reaches the lower end position. You can configure the hold open time during setting door limits (chapter Setting door limits with display) or separately (chapter Setting the hold-open time [1]).
If the door is closed, an Open impulse from the external impulse activator or the Open button on the display initiates the door to open immediately. If the door is open, the Open impulse restarts the hold-open time.
By a Stop command, either external or on the display, the door stops immediately and remains in this position until an operation of the door in the desired direction is initiated by an impulse activator. The display stop can be acknowledged only by an up or down to command on the display, the display stop is not acknowledged by external actuators.
If the door is in the fully opened state, a Stop signal terminates automatic closing and the door remains open until the next impulse. If this is followed by a Close impulse, the closing sequence starts immediately. An Open impulse restarts the hold-open time and initiates the door to close automatically after the time expires.
Interruption of the door line photocell or other safety devices the door reverse to the open position and closes after the hold-open time.

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3.1.2.2 Toggle circuit / manual closing

The toggle circuit / manual closing allows the door to be opened and closed using a single impulse activator. As long as the door is not fully open, every incoming impulse is interpreted as an Open impulse. An impulse received when the door is open starts a closing sequence. For safety reasons, only use manual impulse activators (pull switches or buttons) with this function. Program the input accordingly (chapter Programming the inputs [1]).
Interruption of the door line photocell or other safety devices the door reverses and remain in the open position. Closing of the door is started by a new impulse.

3.1.2.3 Close after passage function

If the close after passage function is activated, the door is opened by an Open impulse activator. The closing sequence is initiated by the interruption of the door line photocell and the door closes after the set hold-open time (Close after passage function). The door closes after the hold open time or after a close impulse if the door line photocell is not interrupted.
The settings for the close after passage function are described in chapter 4.15.

3.1.2.4 Hold to run mode

The door moves as long as the button is pressed and the door has not reached the upper or lower end position. The door switches off automatically when it reaches the upper or lower end position. All safety devices are disabled in hold to run mode.



Observe the safety instructions for "hold to run mode" (chapter Safety instructions for "hold to run mode").



Danger: Dangerous door movement!
Uncontrolled door movement in hold to run mode, if the display unit has a malfunction (e.g. defective foil key buttons).
If the display unit fails, the door movement cannot be stopped in the hold to run mode by means of the display unit.
By pressing the additional emergency stop switch, the door can be stopped safely at any time.

An additional emergency stop switch is required for safe operation of the door. The emergency stop is used to stop the door run in hold-to-run mode in case of a defective display unit. Door operation without this emergency switch is not permitted! This switch must be mounted in the immediate vicinity of the display so that it can be reached directly when operating the display.
For doors without a display unit, the emergency stop switch must be mounted near the display connector (M12 connector) so that the emergency stop is available in the immediate vicinity, when an optional display is connected.

3.1.2.5 Interlocking

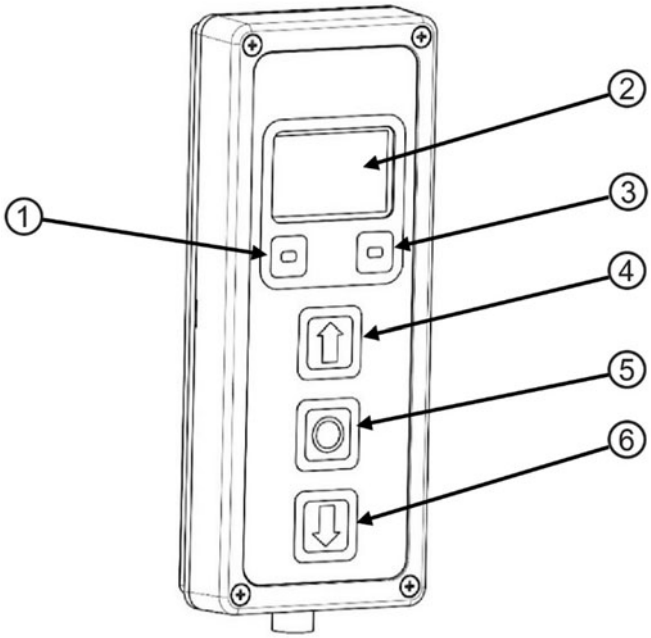
If two or more doors are interlocked, the control units are switched such that only one door can be opened at the same time. If one door is open or opening, all other doors are locked electrically and cannot be opened. The deactivation of the interlocking is described in chapter Deactivation of Interlock function [1].

3.1.2.6 Airlock function

An airlock consists of 2 doors which are automatically controlled by the airlock sequence. Each door has a separate control system. An airlock is used, e.g. to prevent draughts in external doors. The installation of the airlock function is described in chapter Airlock function [1].

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3.2 MCC 2 - Display



1) Function key 1 (F1)
2) Display
3) Function key 2 (F2)
4) Open button
5) Stop button
6) Close button

The three buttons "Open (reset)", "Stop" and "Close" are designed to control the door and to reset the MCC 2 in the case of a malfunction. They can also be used to alter the individual settings (e.g. the hold-open time, etc.). The function of the two function keys is shown on the display at all times. The display is not mandatory for operating the door. The general function can be realized through an external 3-pushbutton station. In this case, the additional emergency stop device for the hold-to-run mode (see chapter 3.1.2.4) needs to be mounted next to the 3-pushbutton station.

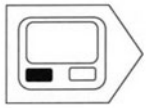
3.2.1 Use of MCC 2 - Display

The display shows messages, faults, etc. in plain text or graphically. Faults during normal operation will be indicated immediately (chapter Error messages (only for doors with displays)).

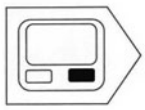
	"Open" button Press the "open" button to send an opening impulse to the control system. In addition, use it to scroll upwards or to increase values in the individual menus.
	"Stop" button Press the "Stop" button to stop the door at any point. The door is locked and can only be operated again, once you press the "Open" or "Close" buttons on the display. The stop function is shown on the display as plain text.
	"Close" button Press the "Close" button to send a closing impulse to the control system. Depending on the test of the safety, there could be a short delay before the closing sequence starts. In addition, use this button to scroll downwards or to decrease values in the individual menus.
	Function keys 1 and 2 The function of these two keys varies and is always shown on the display immediately above the keys.

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The following symbols are used in the instructions to show that the keys should be pressed.



Press left-hand function key!



Press right-hand function key!



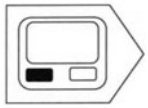
Scroll up or down using the "open" or "close" keys

3.2.2 Save the input

Press the button "Save" to confirm individual inputs. If several digits are changed in a view, these must be saved separately.

3.2.3 Exit the input

Press the left function key "Escape" (up to three times) to terminate the input or parameterisation of the MCC 2 control system with the MCC 2 display:



Alternatively, if no activity occurs, the display returns after a predefined time into the basic view.

3.2.4 Access codes MCC 2 - Display

The MCC 2 display enables the parameterisation of the MCC 2. Due to the ability to change safety-related parameters, the access to the individual parameters is limited by different levels of access codes. Hereby it is possible that there is no access to some parameters.
The necessary access code level is shown in brackets in the headers. There are following options:
No designation = accessible to all
[1] = customer code required (Factory setting 1111)
[2] = installation code required
[3] = system code required

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4. Installation

Check the general operation of the door and the safety devices for proper function after installation of the door.

4.1 Mechanical installation of the control system

Mount the control system with the supplied mounting bracket. You can mount it also onto an oscillation- and vibration-free wall.
The MCC 2 is pre-wired directly to the motor at the factory.

Notice: Risk of material damage!
Improper installation can cause damage to the wiring.
Ensure that the wiring is routed correctly during the control system installation.
When mounting the motor, ensure that the control system is not hanging on the cables.

Close unused cable entries with appropriate sealing plugs. Close any unused plug-in connectors in the junction box with protection cap. Use only cable fittings with at least IP55 rating.

4.2 Electrical installation of the control system

4.2.1 Electrical installation requirements

Danger: Risk of electrical shock!
Safety checks necessary after replacing the internal fuses.
The exchange of internal fuses is only permitted by trained ASSA ABLOY personnel.
Perform a safety check after the replacement.

For proper operation of the control system, the electrical installation must fulfil the following requirements:

- Carry out the electrical work according to the applicable local regulations and requirements.
- The MCC 2 has 3 integrated fuses (dia. 6,3x32mm, 10A, fast Schurter SHF). Locally plan an external 3 pole pre-fuse with 10A - 16A B- or C-characteristic or comparable for the MCC 2. Select the pre-fuse according to the conductor cross-sectional area and the selectivity.
- Additionally observe the national legal regulations!
- Connect the door to the potential equalisation of the building.
- All plugs are marked with "X" on the board.
- An order-specific electrical drawing is included with each door. Wire the control system according to this electrical drawing.
- If a residual current protective device(RCD) is used, an AC/DC sensitive short-time delayed type B RCCB is mandatory. Only this type detects AC residual current, pulsating DC residual current and also smooth DC residual current which could be caused by the MCC 2 in the event of an error. In most European countries RCCBs with a rated residual current of max. 30 mA are demanded. The use of standard RCCBs (type A) is not permissible.
- Additionally, observe the national legal regulations.
  If you are in any doubt please consult your electrician.
- When connecting the shielded cables (motor, brake and encoder cable) be ensured that there is good contact between the shield and the metal cable gland. Otherwise the door could malfunction!
- The main supply cable must have a cross-sectional area of at least 1.5 mm^2. Conductors with cross-sectional areas up to 2.5 mm^2 can be connected.

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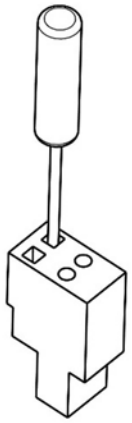
4.2.2 Cross-section wiring / plug-in terminals

Cross-section plug X1, X2, X3, X26, X28 and protective earth terminals:
0.5-2.5 mm² without wire end sleeve, 0.5-1.5 mm² with wire end sleeve without/with plastic collar to DIN 46228-1/-4

Cross-section plug X4-X20
0.14-1.5 mm² without wire end sleeve, 0.14-1.0 mm² with wire end sleeve without/ with plastic collar to DIN 46228-1/-4

4.2.3 Handling recommendation terminal blocks

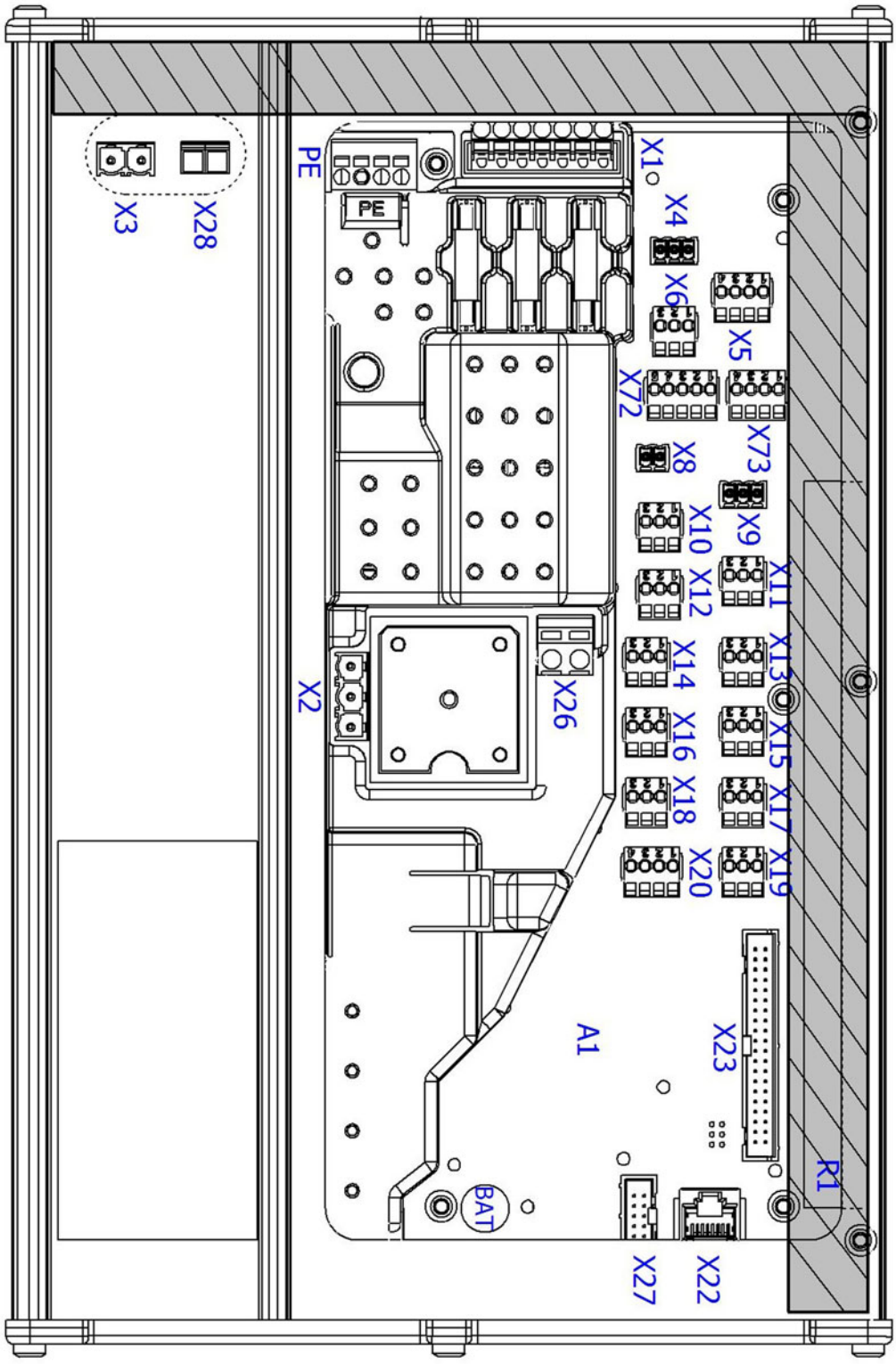
1) Insert a mini screwdriver (slotted 0.4 x 2.5 mm) in the angular operating slot up to the stop.
2) The screwdriver blade holds the clamping spring open automatically so that the conductor can be introduced into the terminal block.
3) Insert the wire into the terminal block.
4) Withdraw the screwdriver – the conductor is automatically clamped.



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4.3 Wiring recommendation and terminal assignment

Install the 24 V wiring in the hatched, grey colored area. Bundle the wires if necessary.



MCC 2 with wiring recommendation

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Warning: Dangerous, not detected failures of the safety devices can occur!

Emergency stop or other external safety-related signals don't fulfill any safety requirements if not connected to the terminals X 72 and / or X 73.
Connect emergency stop or other external safety-related signals to the terminals X 72 or X 73.

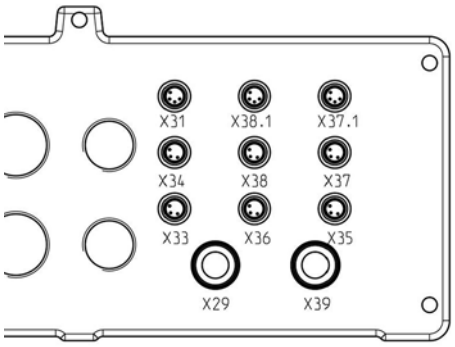
Terminal	Function	Pin assignment
PE	Protective earth connector main voltage and motor cable	
X 1	Main power	1 = L1, 4 = L2, 7 = L3
X 2	Motor connection	1 = U1, 3 = V1, 5 = W1
X 3	Brake	1 = BN, 3 = BU
X 4	Thermal contact	1 = 24 V DC Test output for thermal contact, 2 = Thermal contact, 3 = 0 V
X 5	Encoder	1 = 24V DC, 2 = 0 V, 3 = channel A, 4 = channel B
X 6	Reference limit switch	1 = 24 V DC, 2 = reference limit switch, 3 = 0 V
X 72	Emergency stop 1 (2 channel)	1 = 24 V DC with test pulses for input 1A, 2 = 24 V DC with test pulses for input 1B, 3 = Input 1A, 4 = 0V, 5 = Input 1B
X 73	Emergency stop 2 (OSSD)	1 = 24 V DC, 2 = input channel 2A, 3 = 0 V, 4 = input channel 2B
X 8	Emergency manual lever/hand crank	1 = Input, 2 = 24 V DC Test output for emergency lever
X 9	Receiver, door line photocell	1 = 24 V DC, 2 = photocell, 3 = 0 V
X 10	Transmitter, door line photocell	1 = 24 V DC with Test, 2 = not used, 3 = 0 V
X 11	Safety edge 8,2 kΩ	2/3 = 8,2 kΩ
	Receiver, pre-running safety photocell	1 = 24 V DC, 2 = photocell, 3 = 0 V
X 12	Transmitter pre-running safety photocell	1 = 24 V DC with Test, 2 = Input, 3 = 0 V
X12-X19	General I/O	1 = 24 V DC or output 2 = Input, 3 = 0V
X 20	3-pushbutton station (doors without display)	1 = 24 V DC, 2 = open, 3 = stop, 4 = close
X 22	Modbus interface for display and other operating controls and auxiliary modules	
X 23	M8-junction box and MCC 2 extension box	
X 26	Break resistor	1/3 = 180 Ω
X 27	Clone function	
X 28	PFC Filter connection	1 = L_PFC_1, 2 = L_PFC_2

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4.4 Flange plate with integrated junction box (only for some doortype)

The flange plate with integrated junction box allows an easy and time-saving wiring of the door sensors to the MCC 2. Used only in combination with motor cable length of 1m.

4.4.1 Connector assignment junction box



Junction box

If a safety edge is connected, it may be necessary to use a cable adapter (see electrical drawing)

Danger: Serious injuries at the unsecured door. Loss of the CE conformity.

The inputs and outputs of the junction box and the MCC 2 are interconnected parallel. A simultaneous use can lead to a loss in safety and to unexpected behaviour of the door.
Do not wire/use parallel the corresponding in- and outputs of the junction box and the MCC 2.
Observe the electrical drawing of the door.
Simultaneous use according to order-related electrical drawing is allowed.

Terminal of junction box	Description
X 29	Encoder
X 31	Thermal contact/hand lever
X 33	Reference limit switch
X 34	Reserve I/O 5
X 35	Receiver, door line photocell
X 36	Transmitter, door line photocell
X 39	Display
At doors without crash:	
X 37	Receiver pre-running safety photocell or safety edge
X 38	Transmitter pre-running safety photocell
At doors with crash:	
X 37.1	Crash and receiver pre-running safety photocell
X 38.1	Crash and transmitter pre-running safety photocell

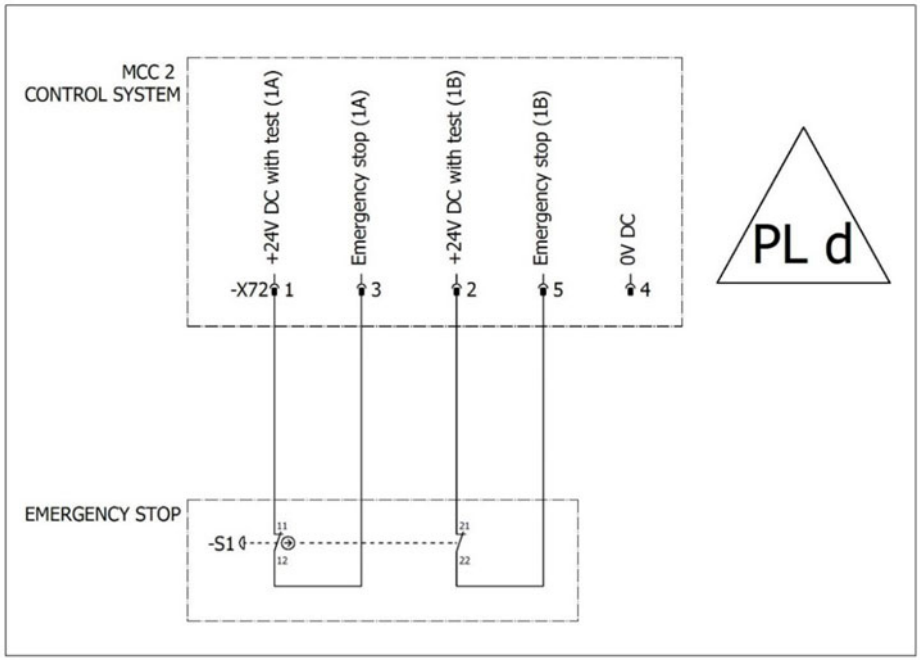
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4.5 Emergency stop input 1 (X72)

The emergency stop input 1 (X72) is a 2-channel emergency stop input with safe torque off function (STO).
The highest safety category (PL d) can only be achieved if the complete emergency stop circuit is performed in 2-channel technology.

Warning: Risk of injury!
A one channel emergency stop circuit doesn't fulfil the requirement for a PL d safety system and has no safety classification.
Use the emergency stop circuit only in two-channel (redundant) design.

For classifications of safety circuits to EN ISO 13849-1 see chapter Classifications of safety circuits to EN ISO 13849-1.
A short circuit between 24 V DC and one input circuit is detected as a fault by the control system.
Sample application note:



2-channel emergency stop on X72

The supplied emergency stop switch to ensure the safe operation in hold-to-run (observe requirement in chapter 3.1.2.4) must always be connected to input X72 with two channels in accordance with safety category 3. The wiring can be found in the order-related wiring diagram.

If a second emergency stop switch is required on site, it must be connected in series with the first one.

When integrating the door into a system, the emergency stop safety concept of the entire system must always be taken into account. If the safety concept requires it, the emergency stop must also be integrated into the system control in a safety-related manner. In this case, the system control must then activate the MCC safety input X72 to ensure a safe stop of the door. The entire emergency stop safety chain must have two channels (Cat. 3).
The mechanical installation of the emergency stop switch including holder is described in the installation instructions of the door. The connection needs to be done on site.

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4.6 OSSD Emergency stop input 2 (X73)

This input is designed for connection of safety units with 2 OSSD outputs like laser scanner, light curtain or similar safety components.
The emergency stop input 2 (X73) is a 2-channel OSSD emergency stop input with safe torque off function (STO). The highest safety category (PL d) can only be achieved if the complete emergency stop circuit is performed in 2-channel technology.

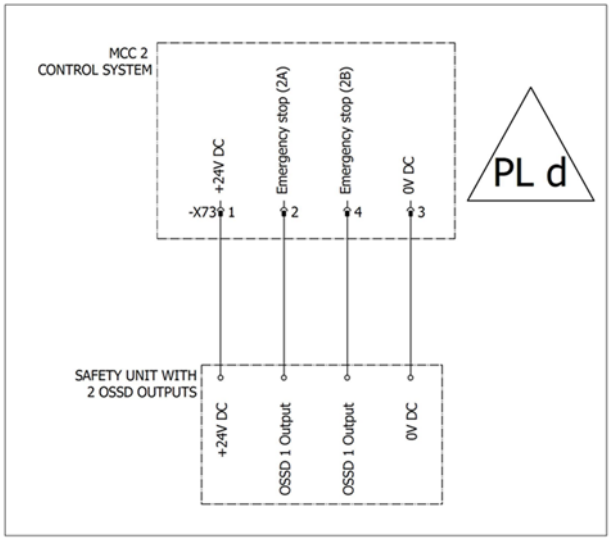
Warning: Risk of injury!

A single channel emergency stop circuit doesn't fulfill the requirement for a PL d safety system and has no safety classification.
Use the emergency stop circuit only in two-channel (redundant) design.

Warning: Risk of injury!

An external emergency stop unit without OSSD pulses doesn't fulfill the requirement for a PL d safety system and has no safety classification.
Use the emergency stop circuit only with OSSD outputs in two-channel (redundant) design.

For classifications of safety circuits to EN ISO 13849-1 see chapter Classifications of safety circuits to EN ISO 13849-1.
A cross fault between both inputs on the external cable should be detected by the OSSD output sensor itself. The control system detects cross faults inside the electronic as a fault by the control system during the regular test.
A short circuit between 24 V DC and one input circuit is detected as a fault by the control system.
Sample application note:



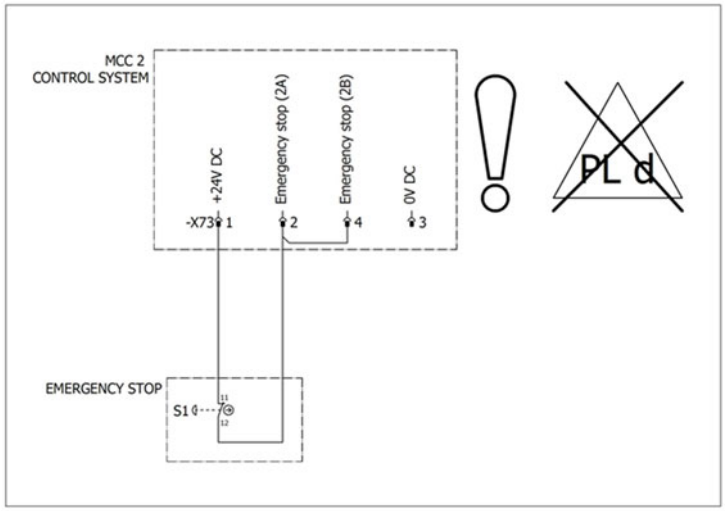
2-channel OSSD emergency stop on X73

4.6.1 Connection of a single channel sensor to X73:

The highest safety category (PL d) can only be achieved if the complete emergency stop circuit is performed in 2-channel technology.

Warning: Risk of injury!
A single channel emergency stop circuit doesn't fulfil the requirement for a PL d safety system and has no safety classification.
Use the emergency stop circuit only in 2-channel (redundant) design.

For classifications of safety circuits to EN ISO 13849-1 see chapter Classifications of safety circuits to EN ISO 13849-1.
Cross faults between both input circuits and short circuits between 24 V DC and one input circuit also will be not detected as a fault by the control system.
Sample application note:



1-channel emergency stop on X73

4.7 Calculation of the complete system safety values

With the safety value from the external sensor and the safety value from the MCC 2 control system input you can calculate the complete system safety value.

The following formulas are valid only for the assumption that the MTTFd values for both channels of the redundant input are the same.

Total MTTFd of the input circuit with sensor:

$$\frac{1}{MTTF_{d\,tot}} = \frac{1}{MTTF_{d1}} + \frac{1}{MTTF_{d2}}$$

MTTFdtot = Total MTTFd value of the complete input circuit with external sensor
MTTFd1 = MTTFd of the external safety sensor
MTTFd2 = MTTFd of the MCC 2 control system input (value see chapter Classifications of safety circuits to EN ISO 13849-1)
Total DCavg of the input circuit with sensor:

$$DC_{avg} = \frac{\frac{DC_1}{MTTF_{d1}} + \frac{DC_2}{MTTF_{d2}}}{\frac{1}{MTTF_{d1}} + \frac{1}{MTTF_{d2}}}$$

DCAVG = Total DCAVG value of the complete input circuit with external sensor

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DC1 = DCAVG of the sensor
DC2 = DCAVG of the MCC 2 control system input (value see chapter Classifications of safety circuits to EN ISO 13849-1)
MTTFd1 = MTTFd of the external safety sensor
MTTFd2 = MTTFd of the MCC 2 control system input (value see chapter Classifications of safety circuits to EN ISO 13849-1)

4.8 Reference limit switch

All ASSA ABLOY Doors are supplied with a reference limit switch mounted in the upper end position. This reference limit switch indicates the zero (reference) point for door positioning via the encoder. There are 2 types of reference limit switches:

- Mechanical spindle or cam reference limit switch. This type of limit switch must be set to the upper end position mechanically during installation (see also User Manual for the door).
- Inductive reference limit switches. This type of limit switch needs to be adjusted during installation.

4.9 Setting door limits with display

Set the end positions before using the door.



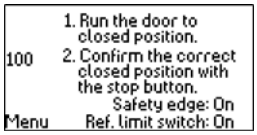
Notice: Risk of material damage!

If the door is moved beyond the lower end position manually during installation (in hold to run mode), this may result in severe damage to the door.
Observe the door during setting the end positions permanently.



While setting the door limits the door will only move in creep mode in hold to run mode.

The control system is delivered without the limit positions set; therefore, this view will be the first that appears when the control system is powered up. You can also set the door limit positions by pressing the open and stop buttons on the display together for approximately 10 seconds. The following message appears:



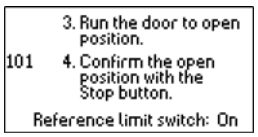
In this view the setup cannot be restarted again through simultaneous pressing of the Open and Stop button. For renewed activation of the setup, press the Stop pushbutton and then restart the setup by simultaneously pressing the Open and Stop button or by switching the Main switch off and back on again.



The Menu button can be used to quit the setup at this point to check or change the default settings (e.g. brake voltage).

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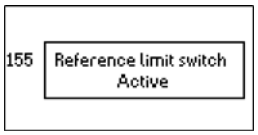
Now use the display to move the door to the lower final position and save the position by pressing the stop button on the display (If the door is already in the closed position, the position can be saved without moving the door.). The following message appears:



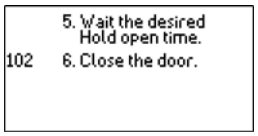
Notice: Risk of damage to the door because of incorrectly set reference limit switch!

Check whether the reference limit switch needs to be readjusted.

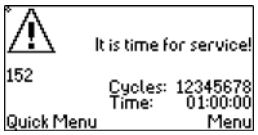
For doors with a reference limit switch, use the display to move the door to the upper end position until it stops automatically. The following message appears:



Save the upper end position by pressing the stop button on the display. The following message appears:



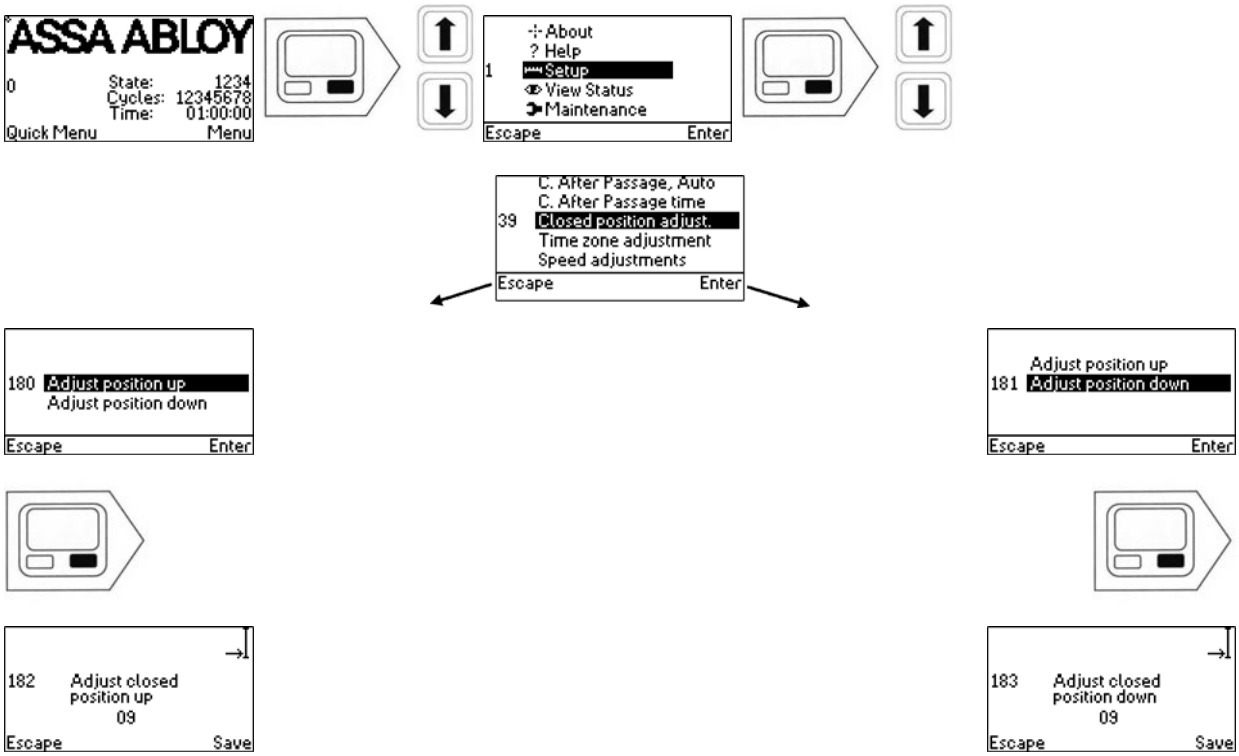
(Be ready for this step it starts once the top position is set!)
The hold-open time can now be set. For this wait for as long as the door should remain open in future. To save this press the "close" button on the display.
Example:
Hold-open time = 6 seconds => after saving the upper end position wait for 6 seconds and then press the "close" button. If you press the close button within 2 seconds, the hold-open time is switched off. Instead of using the Close button, you can also deactivate automatic closing (hold open time = 0) by pressing the Open button (chapter Setting the hold-open time [1]).
After the 1st setup you must reset the service interval (chapter Reset service interval [1]), otherwise the following message appears after several cycles.



This resetting can be omitted in subsequent end position programming.

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4.10 Correcting the lower end position [1]

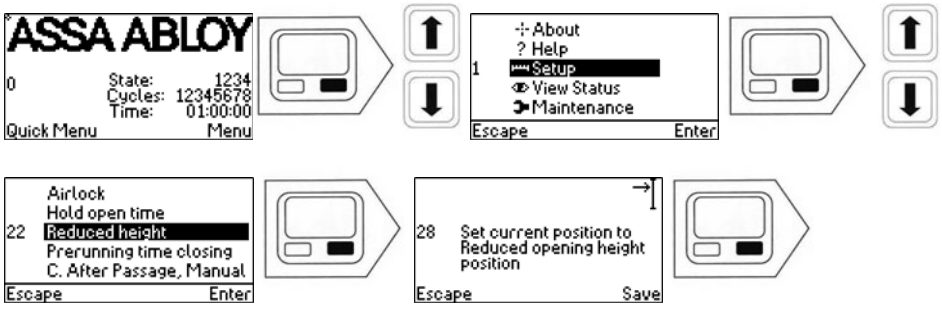


To make sure that the lower end position is not passed, only change the limit in small steps! 99 pulses are approximately 1-2 cm (0,39"-0,78"). This depends on the door type.

4.11 Setting the reduced opening height [1]

The corresponding MCC 2 input, which should open the door to the reduced opening height, must be programmed to the function "Reduced opening height".

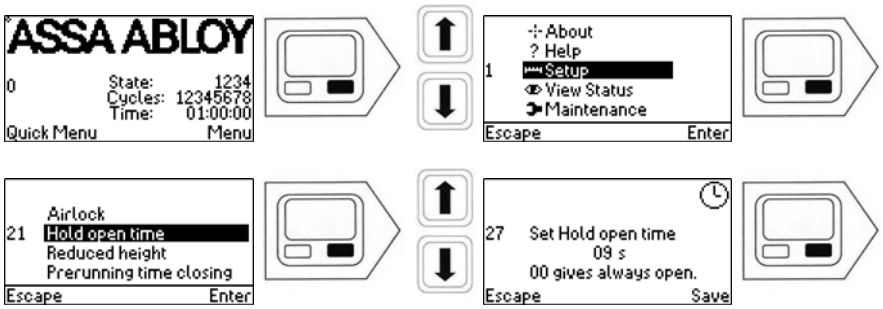
Move the door to the reduced opening height desired. Save the position as follows:



The reduced opening height is only used for inputs with the appropriately programmed function!

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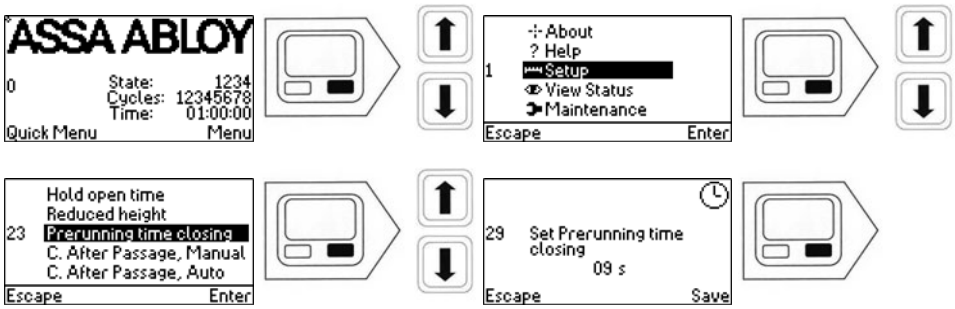
4.12 Setting the hold-open time [1]



This setting is also part of the quick menu.

4.13 Setting the pre-running time before closing [1]

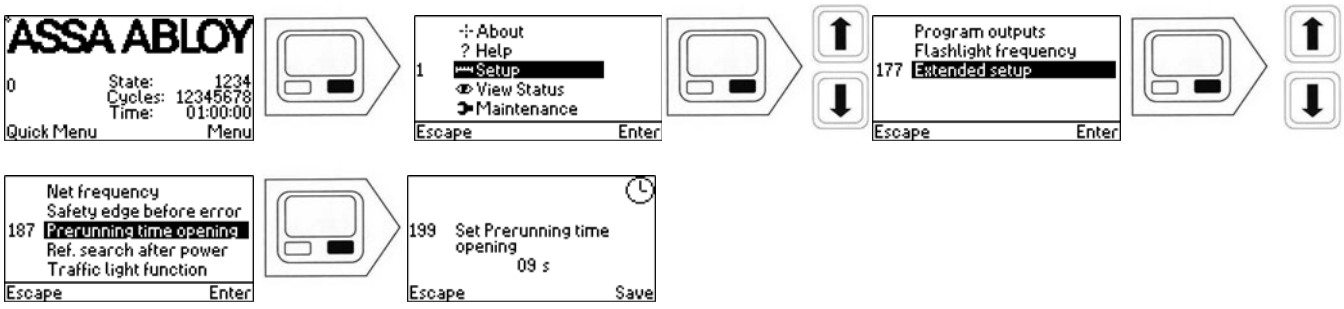
This function is used for traffic light, horn etc. as a warning signal before the door is closing.



This setting is also part of the quick menu.

4.14 Setting the pre-running time before opening [1]

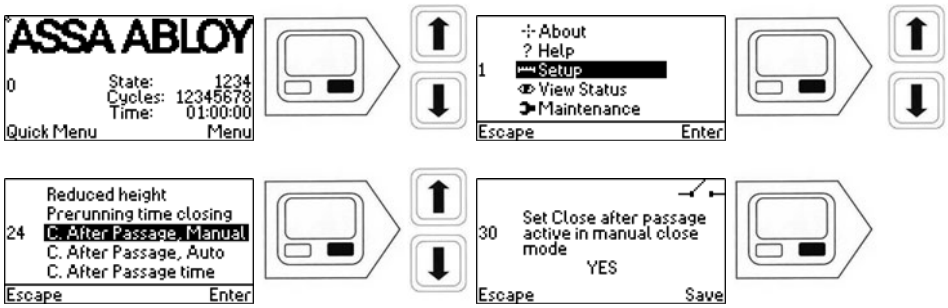
When this pre-running time is activated there will be a delay before the door opens (it opens after the pre-running time).



4.15 Close after passage function

4.15.1 Setting the close after passage function without hold-open time [1]

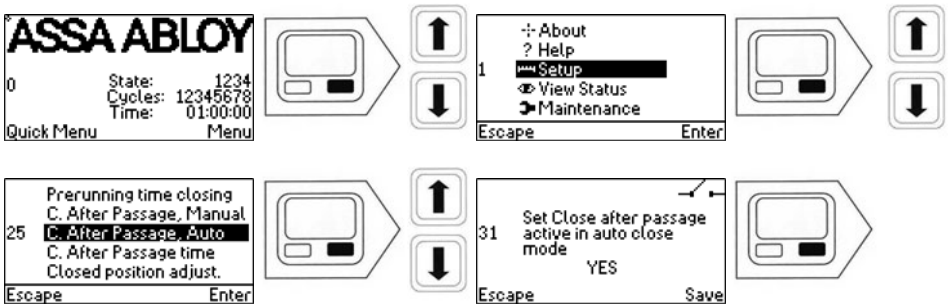
Use this setting to set the close after passage function at actuator inputs with the function "Open command without automatic close function".



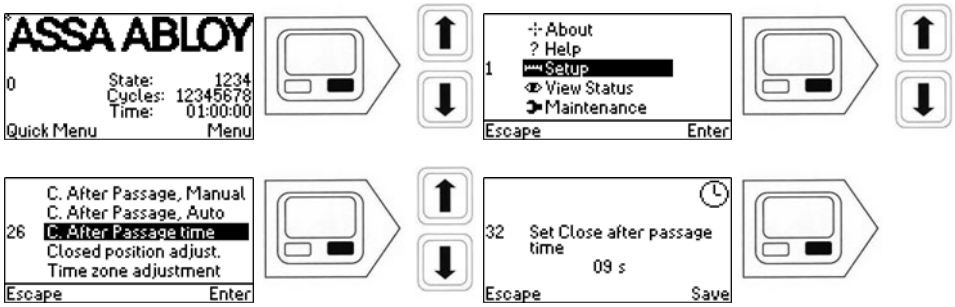
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4.15.2 Setting the close after passage function with hold-open time [1]

Use this setting to set the close after passage function at actuator inputs with the function "Open command with closing after time".



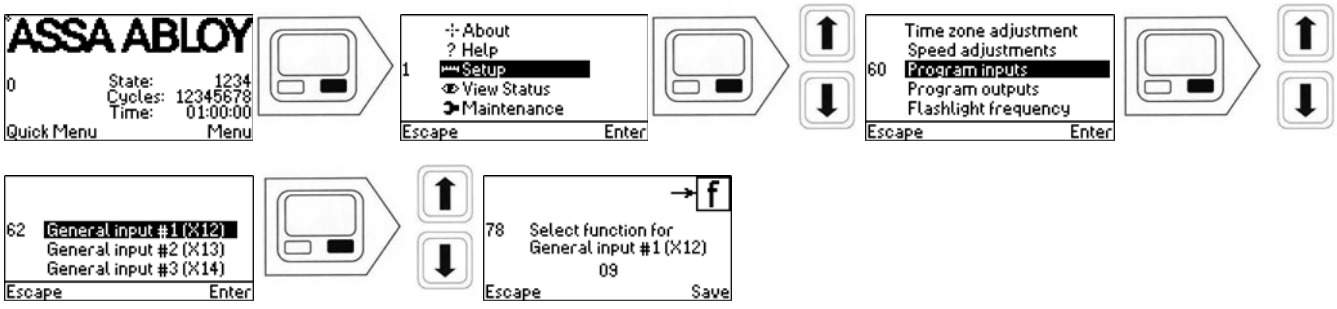
4.15.3 Setting the close after passage time [1]



4.16 Programming the inputs [1]

Select the input to be programmed (observe the electrical drawing) and enter one of the function codes listed below.

For safety reasons inputs 1 and 2 (X12 and X13) can only be re-programmed with system code [3] by the ASSA ABLOY Entrance Systems Service Team.



For traffic lights with recognition of direction it is important that the impulse activators are assigned to the direction. The front is always the side of the door where the display / foil key-pad panel is mounted. If direction dependant traffic lights are not used only "front" opening functions shall be used on both sides of the door.
The options are as follows:

Code No	Function
41	Open command with closure after a time (front)
42	Open command with closure after a time (back)
43	Open command (reduced opening height) without automatic closing function

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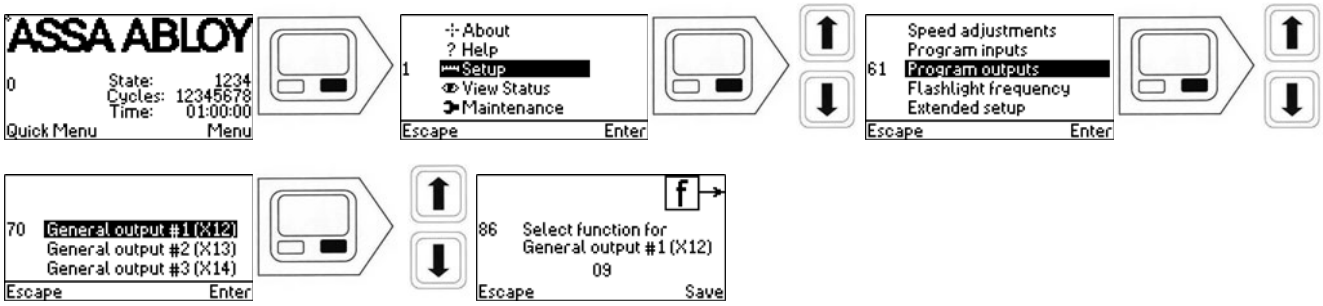
Code No	Function
44	Open command (reduced opening height) with automatic closing function (front)
75	Open command (reduced opening height) with automatic closing function (back)
22	Input without function (e.g. deactivating actuators)
23	Open command (normally open) without automatic closing function
24	Close command (normally open)
25	Stop command (normally closed) -> Use Input X72 for emergency stop!
26	Toggle circuit
69	Additional safety (normally closed)
51	Additional safety (normally open)
53	Interlock input (normally closed)
80	Interlock input (normally open)
60	Interlock (switch on/off)
56	Lock door, e.g. mech. door locking, pass door, etc. (normally closed)
54	Crash (normally closed) (function only possible on X12 and X13!)
57	Long vehicle input for airlock (normally open)
45	Middle actuator input for airlock (normally open)
58	Permanent opening display
59	External permanent opening
91	PLC Input 1
92	PLC Input 2
93	PLC Input 3
94	PLC Input 4
95	PLC Input 5
96	PLC Input 6
97	PLC Input 7
98	PLC Input 8
62	Open command (normally closed contact) without automatic close function
71	Breakaway (normally closed contact) (only for doors with breakaway function)
67	Input low-battery waring wireless safety edge system

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4.17 Programming the outputs [1]

Select the output to be programmed (electrical drawing) and enter one of the function codes listed below.

For safety reasons outputs 1 and 2 (X12 and X13) can only be re-programmed with system code [3] by the ASSA ABLOY Entrance Systems Service Team.



The options are as follows:

Code No	Function
65	+24 V DC (continuous voltage)
28	Traffic light (red/green), front (activation relay)
29	Traffic light (red/green), back (activation relay)
30	Switch traffic light on/off when door closed
76	Traffic light green 24 V DC front
77	Traffic light green 24 V DC back
78	Traffic light red 24 V DC front
79	Traffic light red 24 V DC back
63	Output without function (e.g. deactivating activators)
31	Flashlights
33	Interlock
34	Door open
39	Door open to reduce height
35	Door closed
36	Door ready for work
81	Malfunction indication
37	Status light
38	Pre-running safety photocell transmitter
40	Additional safety
83	PLC output 1
84	PLC output 2
85	PLC output 3
86	PLC output 4
87	PLC output 5
88	PLC output 6

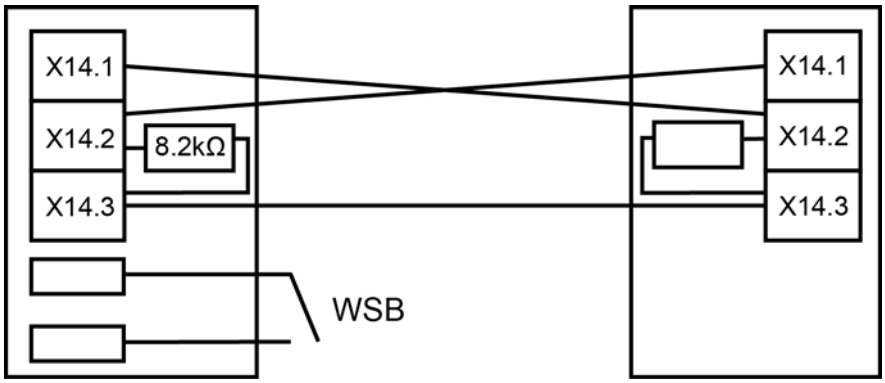
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Code No	Function
89	PLC output 7
90	PLC output 8
72	UPS On/Off control

4.18 Airlock function [1]

4.18.1 Airlock wiring

A 3-wire connecting cable and two 8.2kΩ resistors are required for the airlock lock between door A and B. Two 8,2 kΩ resistors are also necessary. The airlock connection of both MCC 2´s is only possible at the terminals X14!
The following connections are to be provided (Observe electrical drawing of the door!):

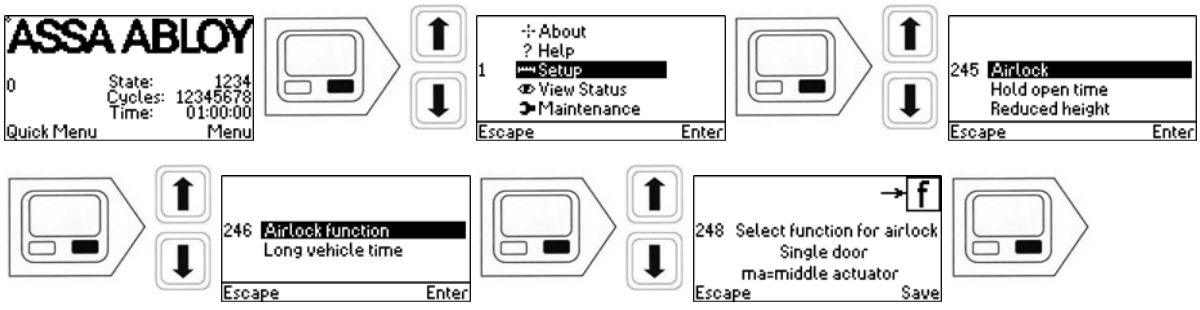


The door with the middle actuator (WSB) is always Door A! The free inputs X12-13, X15-19 can be used.

4.18.2 Setting the airlock function [1]

The functions of the X14 input must be programmed to code number 22 (input without function). The output X14 must be programmed to code number 65 (24 V continuous voltage).

In this menu item select whether a single door, Door A or B of the airlock is concerned and whether the airlock is operated with or without middle actuator (WSB).
You can activate the close after passage time for actuators with hold open time and for actuators without hold open time (chapter Setting the close after passage function without hold-open time [1] / Setting the close after passage function with hold-open time [1]).



The selection options are:

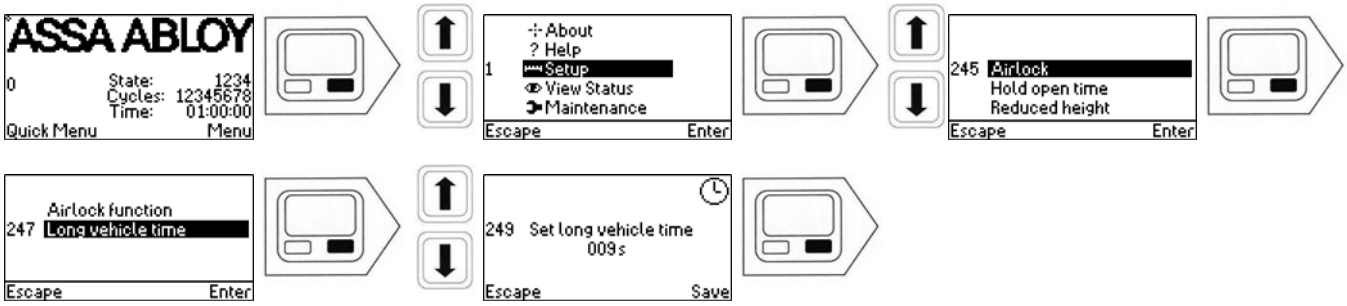
Function	Description
Single door	No airlock operation

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Function	Description
Airlock with middle actuator Door A	Airlock operation with middle actuator at Door A (middle actuator is connected to door A)
Airlock with middle actuator Door B	Airlock operation with middle actuator Door B (middle actuator is connected to door A)
Airlock without middle actuator Door A	Airlock operation without middle actuator door (A)
Airlock without middle actuator Door B	Airlock operation without middle actuator door (B)

4.18.3 Setting the long vehicle time [1]

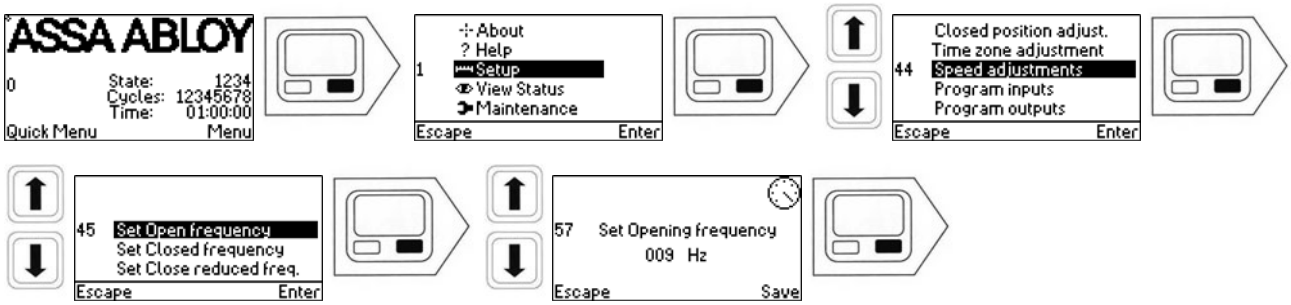
The long vehicle time is the time which the door line photocell in the side frame and the middle actuator have to be simultaneously activated to start the long vehicle function. With the long vehicle function both doors open regardless of the airlock sequence. The factory setting is 15 s.



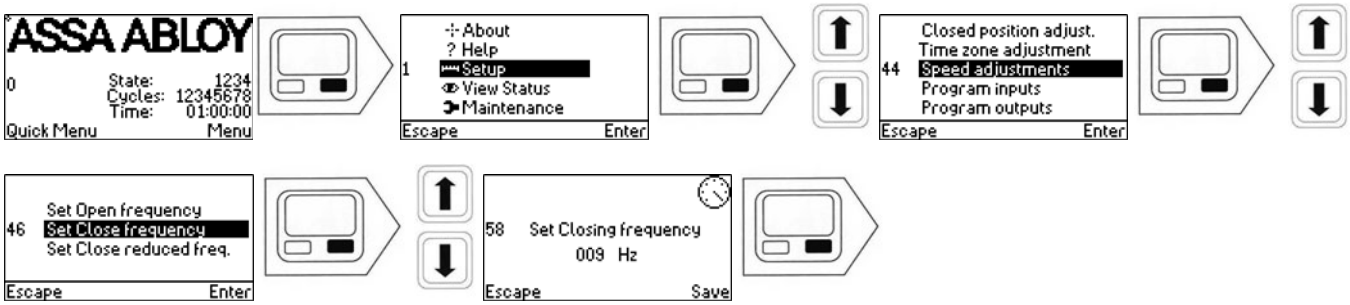
4.19 Reducing the speed [2]

For safety reasons the speeds can only be altered within a pre-defined range.

4.19.1 Reducing the opening speed [2]

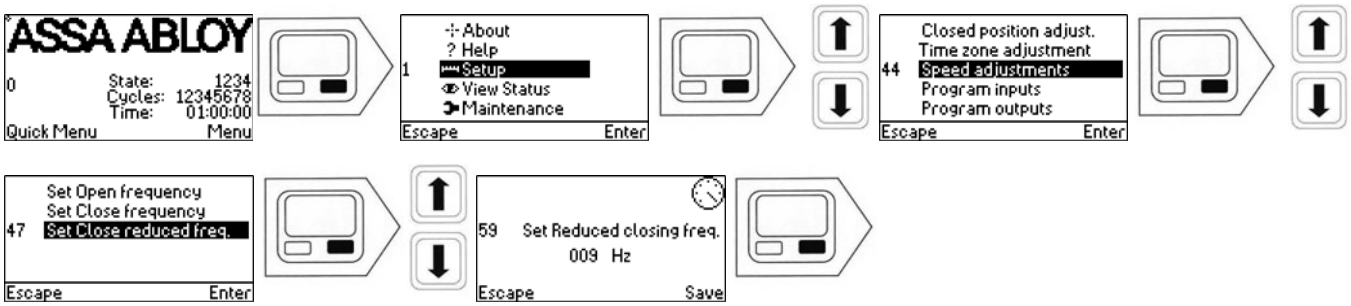


4.19.2 Reducing the closing speed above an opening height of 2.5 m [2]



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4.19.3 Reducing the closing speed below an opening height of 2.5 m [2]

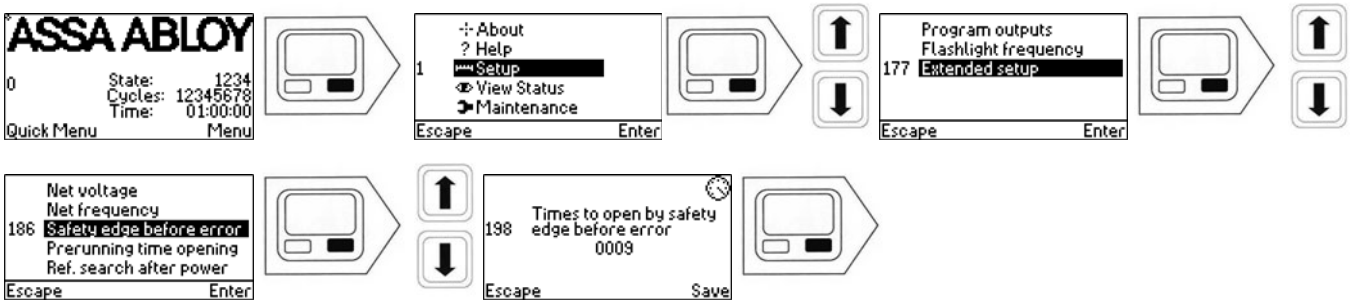


4.20 Adjustment of activations of the safety edge or pre-running photocell [2]

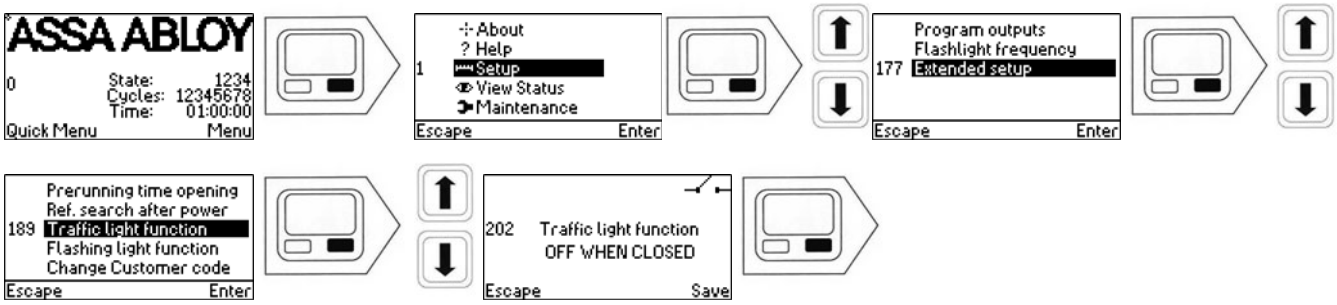
The electrical safety edge serves as closing edge protection. As standard, the door is usually locked in the upper end position after the electrical safety edge has activated once. Using this menu, you can increase the value, for example to twice. This means that the door would only lock in the upper end position after two activations of the electrical safety edge (without the door closing completely). For the pre-running safety photocell this value is set to 10 activations as standard.

Warning: Personal risk at the door is increased!

A collision with the door curtain can be repeatedly performed with increasing the value.
The value should only be altered by our service team in reasoned, exceptional cases.
For safety reasons the value for the electrical safety edge should only be increased to a maximum of 3 activations!



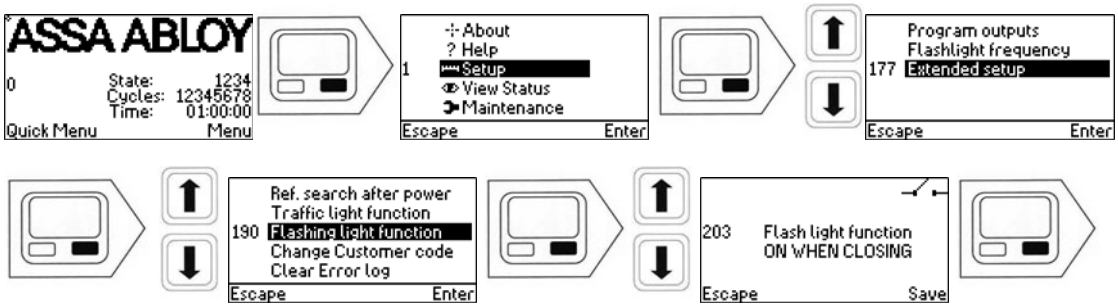
4.21 Altering the traffic light function [1]



Select whether the traffic light remains on all the time or switched off when the door is closed.

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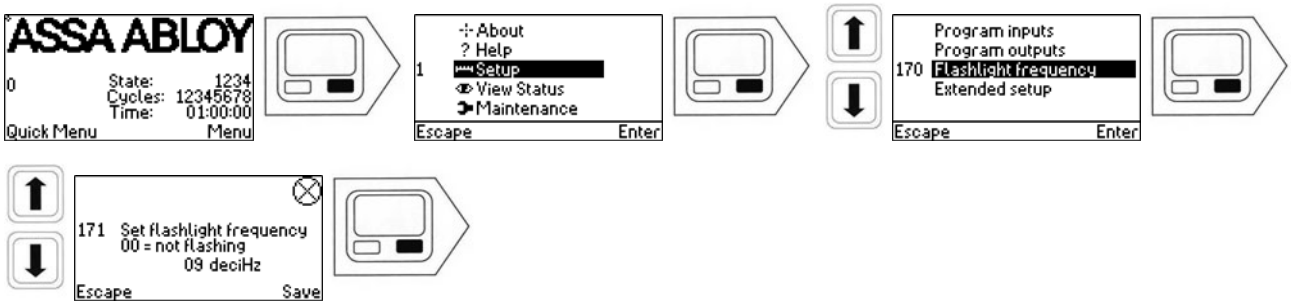
4.22 Altering the flashing light function [1]



Select whether the flashing light is on whenever the door moves or only during closing.

4.23 Setting the flash rate of the flashlight [1]

The control system offers the option of using a normal indicator light as a flashlight. Change the switching frequency of the flashlight using the following menu option:

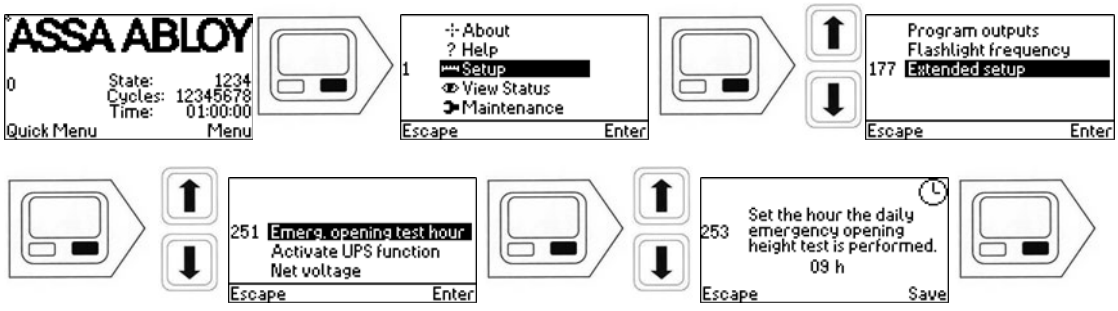


4.24 Emergency exit doors – set time for regular self-test [1]

To ensure the emergency exit and escape route functions, self-tests are performed by the control system after switching on, after the escape route deactivation has ended, after 24 hours and after a power failure.

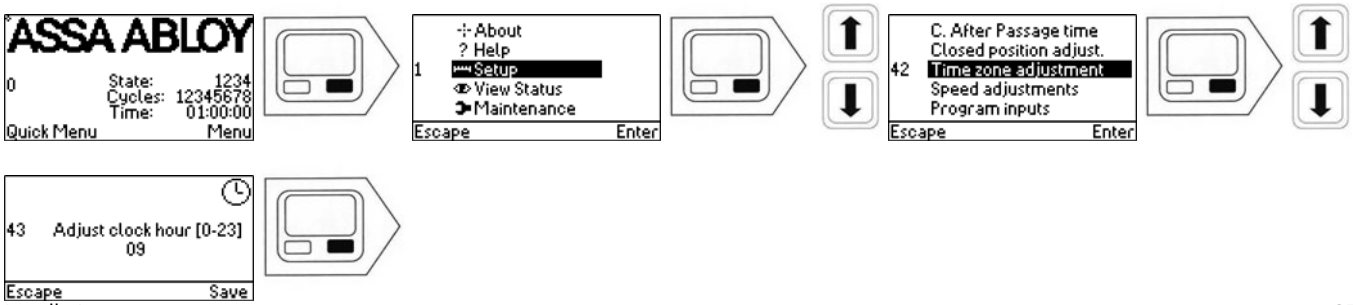
When self-tests are performed, all safety relevant electrical and mechanical functions of the door are checked.

The factory setting for the self-tests after 24 hours is 8:00 a.m. Change this time to your need using the display unit.



4.25 Setting the time zone [1]

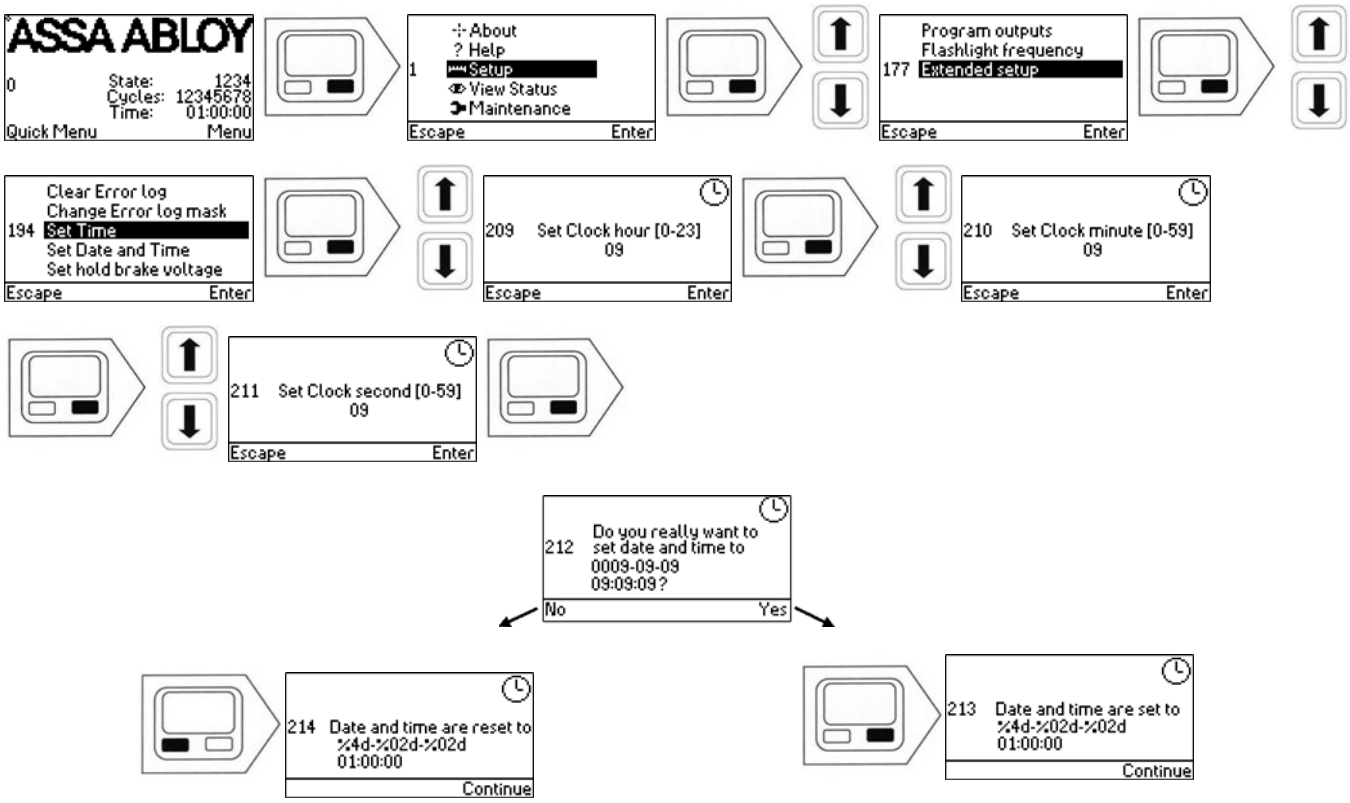
If the door is placed in a different time zone you have to change the clock hour.



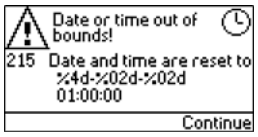
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4.26 Setting of the time [1]

If the clock setting is not correct, you have to change the time.



If the adjusted values are not valid the following view appears:

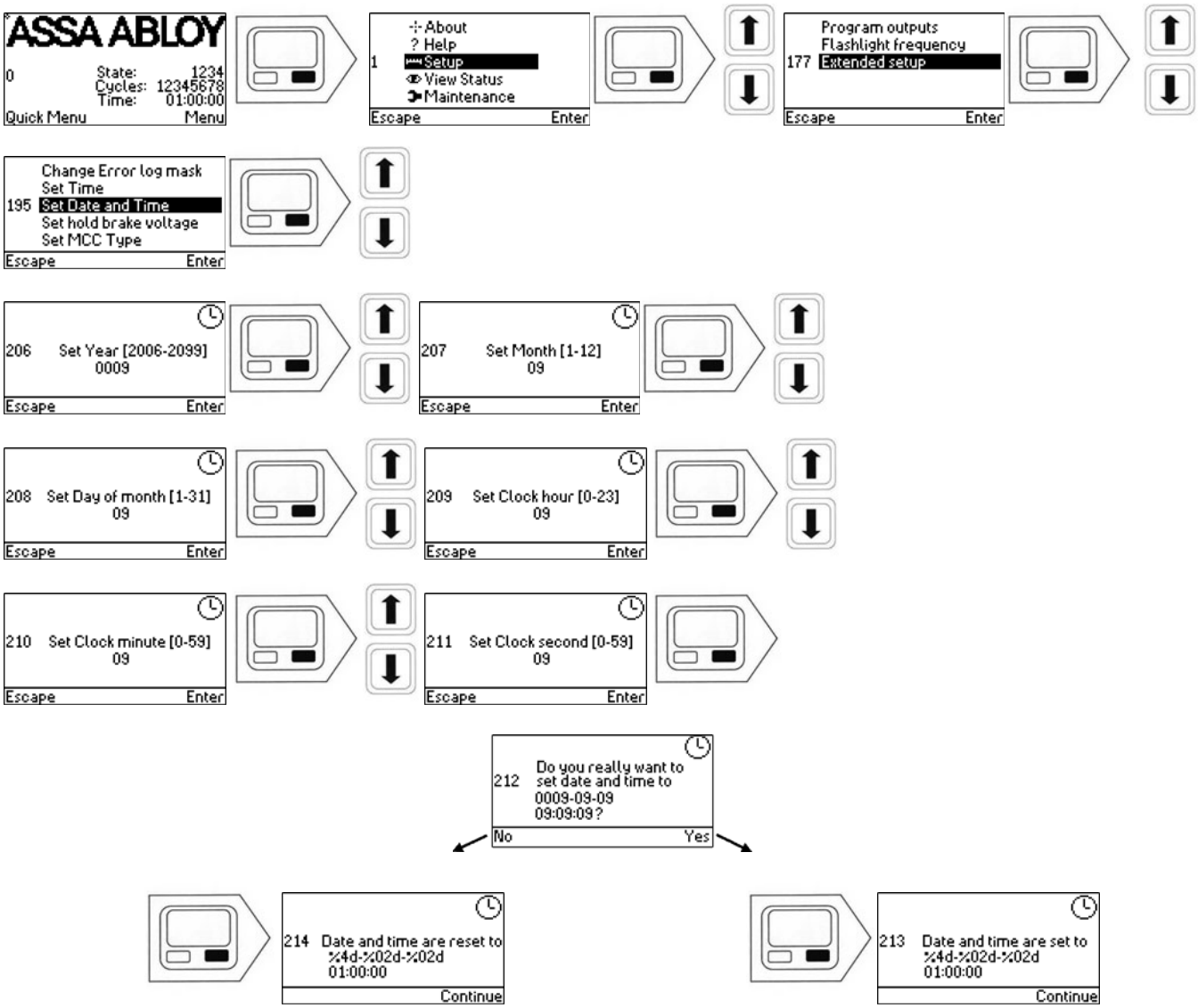


Date and time will be set to the old values.

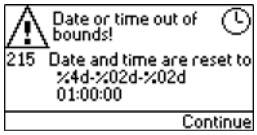
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4.27 Setting of the date and time [2]

Here you can correct the control system date and clock.



If the adjusted values are not valid, the following view appears:

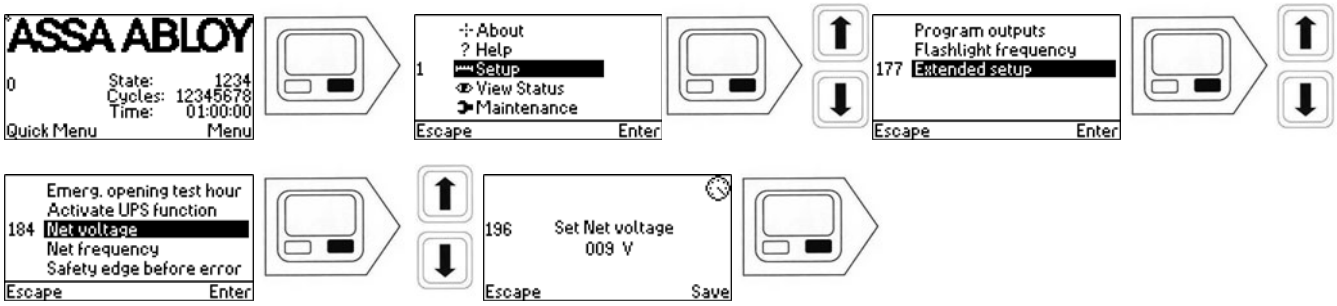


The time will be set to the old values.

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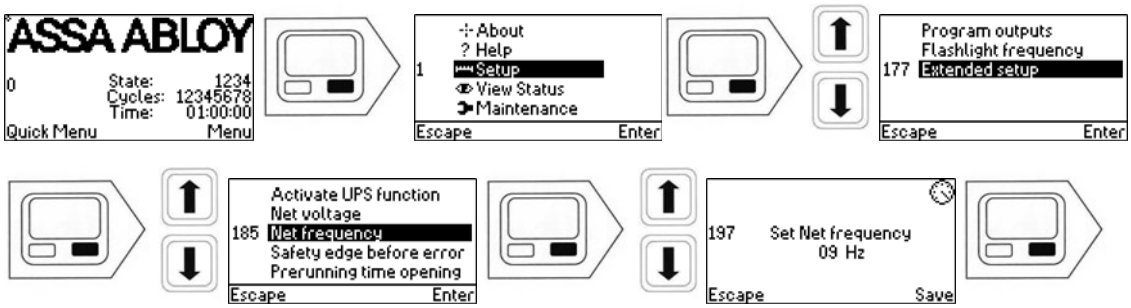
4.28 Changing the main or supply voltage [2]

Notice: Risk of material damage!
A wrong main voltage setting can destroy the control system.
Check local main voltage if necessary and adjust accordingly in the MCC 2.



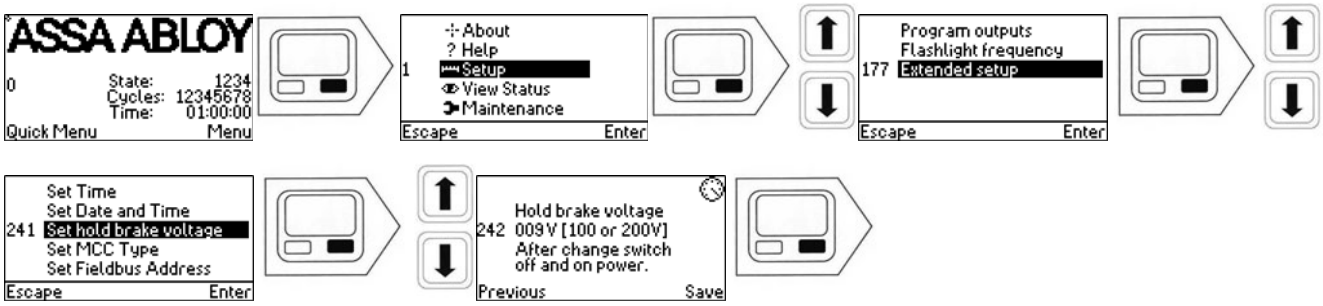
Switch the control system off and on again, so that the main voltage change becomes effective.

4.29 Changing the main supply voltage frequency [2]



4.30 Changing the brake voltage [2]

Notice: Risk of material damage!
A wrong brake voltage setting can destroy the control system and the brake.
Observe the information of the type label of the brake.

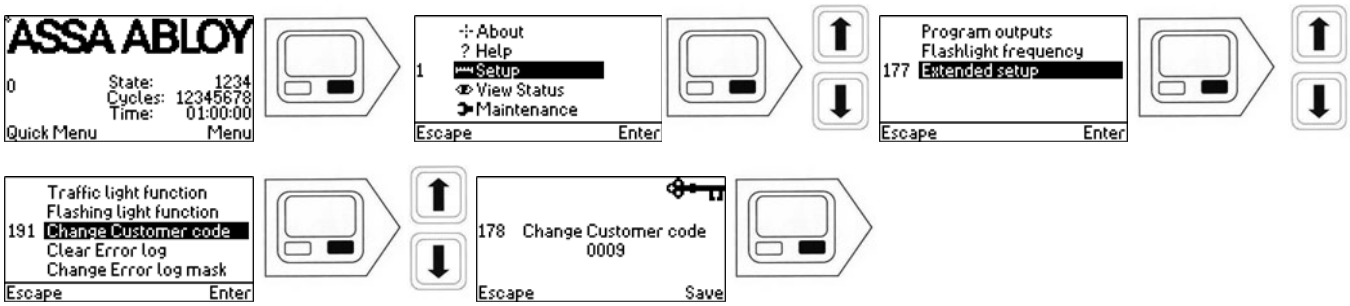


Switch the control system off and on again, so that the brake voltage change becomes effective.

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4.31 Change customer code [2]

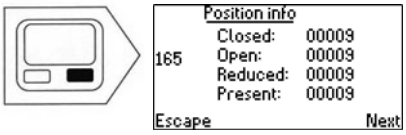
In order to prevent that the user locks their selves out, the code can only be changed by the service technician.



4.32 Querying the door positions and the state history [0]



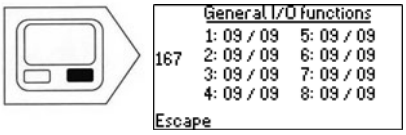
In this view you can read off the set date, set time, current MCC 2 temperature, the current software state and the ready for work signal.



In this view you can read off the number of encoder pulses for the end positions closed, opened and reduced opening height and the current position.



The last 10 and the current state are displayed in this view.

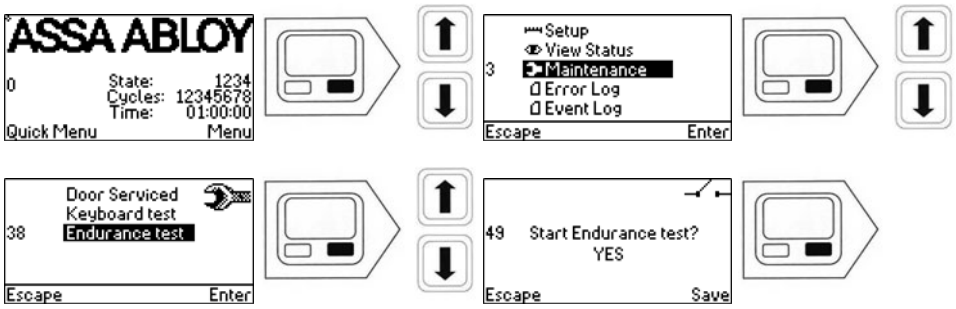


The functions of the programmed inputs and outputs are displayed in this view (for meaning of the codes, see chapter 4.16 / 4.17).

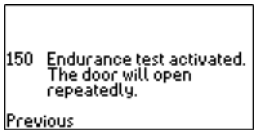
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4.33 Setting endurance test [2]

Only activate this function for test purposes!



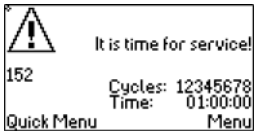
Activated continuous operation is indicated by the following message:



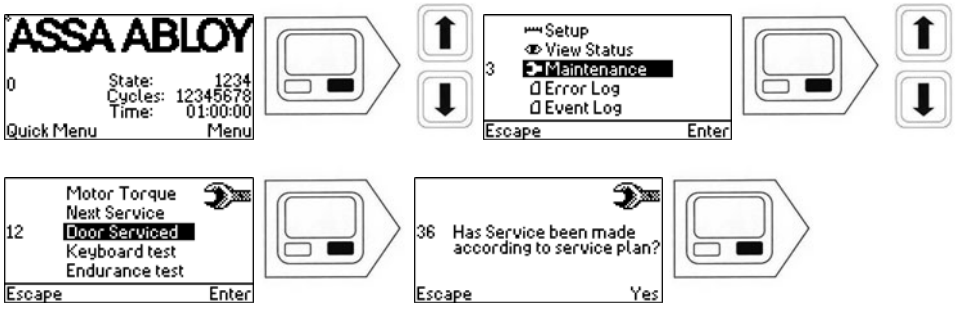
Deactivation is performed in the same order or by switching power off and on again!

4.34 Reset service interval [1]

If service is due, it is automatically shown with the following message.



Every service interval indicated is recorded in the event log.
If the service has been carried out in accordance with ASSA ABLOY Entrance Systems specifications you must reset the service interval counter. Every time the service interval counter is reset this is recorded in the event log.



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5. Operation

5.1 Safety instructions

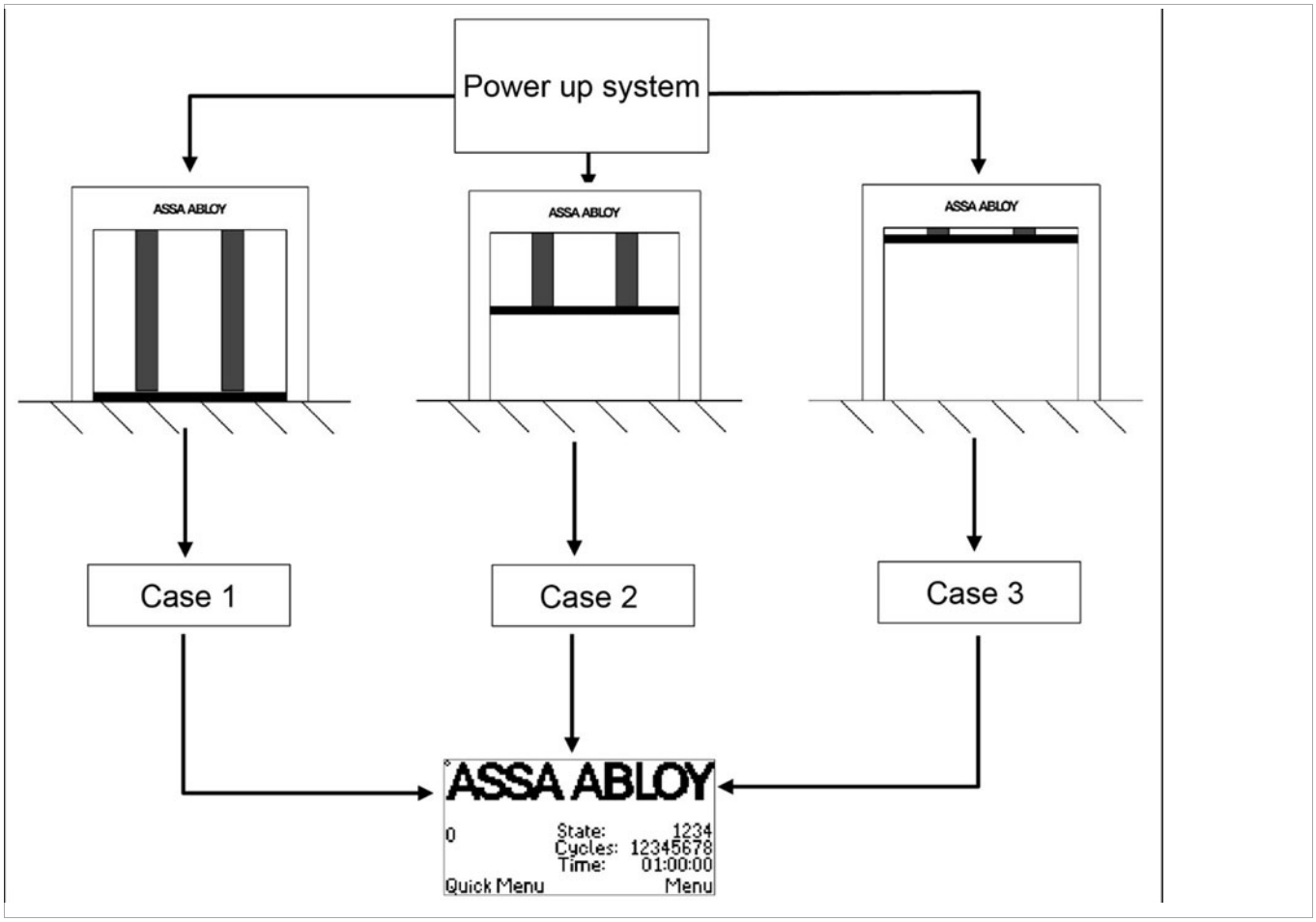
Observe the safety instructions in chapter 2 and the safety instructions in the user manual for the door.

5.2 Display

The display shows messages, faults, etc. in plain text or graphically. During operation the time, cycle counter and status information are shown.

5.3 Switch-on and reference run

The position of the door and the vector control of the motor are regulated by an incremental encoder. This encoder is firmly mounted on the motor. A reference search is therefore necessary when the system is switched on.



Case	Button	Function
1	 or external open command	Door opens completely in slow speed, and closes automatically after an allocated hold open time. If the open time = 0, door closes after next close pulse.

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1	 or external close command	Door opens completely in slow speed and closes automatically directly after open end position has been reached.
2	 or external open command	Hold open completely in slow speed, and closes automatically after an allocated open time. If the hold open time = 0, door closes after next close pulse.
2	 or external close command	Door opens completely in slow speed and closes automatically directly after open end position has been reached.
3	 or external open command	After allocated open time has passed, door closes automatically by 20 cm in slow speed (if hold open time = 0, the door stays open). Then the door stops and opens again into open end position in slow speed (door is completely opened) and then directly closes in automatic mode.
3	 or external close command	Door directly closes automatically by 20 cm in slow speed. Then door stops and opens again into the open end position (door is completely opened) and then directly closes in automatic mode.

The door is now ready for operation.

5.4 Opening the door

An "open" impulse is given by any impulse activator or from the display. The brake is released; the drive motor is activated by the MCC 2 with a pre-defined acceleration ramp. The door opens with maximum speed until it reaches the starting point of the upper deceleration ramp. The door is now decelerated by the frequency converter and stopped in the adjusted upper end position. When this position is reached the motor is switched off and the brake is activated.
Depending on the function, which has been pre-selected, either the hold-open or close after passage time is started or the control system waits for a closing impulse.

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5.5 Closing the door

Once the end of the hold-open time or close after passage time is reached or a closing impulse is received all connected safety devices are first tested by the control system. If this safety test is successful, the brake is released and the motor is driven by the frequency converter with a pre-defined acceleration ramp. The door closes as far as the starting point of the lower deceleration ramp. The door is now decelerated with a pre-defined deceleration ramp and stops in the lower end position. When this position is reached the motor is switched off and the brake is activated.



The safety edge and the pre-running photocell are switched off max. 50 mm (2") before the lower end position is reached. The switching off point for the doorline photocell could be different. This prevents the door from reversing when it reaches the ground.

5.6 Stop the door

The door can be stopped in any position by pressing the "Stop" button. If the stop button on the display has been pressed, this can only be reset by pressing the open or close button on the display. The control system also remembers that it is stopped with the display stop button, if it is restarted by switching on and off the main supply. Also, in this case only the display open or close button can start the door again.

5.7 Power failure

If a power failure occurs, doors with power off brake immediately stops wherever it is at that moment. Once the main power returns the door will only be in reference search mode. Doors with power on brake open or close automatically in the event of power failure depending on the spring balancing and door position. A closed door will open and an open door can close slightly (depending on the spring balancing).

5.8 Emergency manual operation

In the event of a power failure, during repair works or maintenance, it may be necessary to open the door manually.



Doors with power on brake do not have an emergency manual lever!



Warning: Risk of injury by the movement of the door curtain!

The door may partially open or close by itself after the emergency manual lever has been activated!
Observe the door during use of the emergency manual lever permanently.



Warning: Risk of injury!

It can lead to injuries by simultaneous starting of the motor during inserting the emergency hand crank.
Always switch off the main switch before using the emergency hand crank.

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By using the emergency manual lever of the door, you can release the brake of the drive motor via a disengagement cable. Depending on the balancing system and the door position, the door curtain may partially open or close by itself. Reach below the bottom rubber in the middle of the door and push the door curtain up. While doing this the door curtain will wind onto the top roll. To get back to the normal operation of the door, return the emergency manual lever to its non-actuated position. Switch the control system on and operate the door in reference search mode.
At doors with emergency hand crank, open and close the door by turning the emergency hand crank which must be inserted on the bottom of the motor.
If the emergency manual lever or the emergency hand crank is still operated when the main switch is switched on the following message appears:



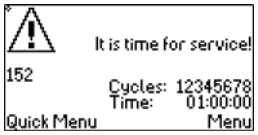
This message is reset automatically once the emergency manual lever has been returned to its original position or the emergency hand crank has been removed. A reference search is needed.
The reference search is initiated by pressing the "Open" button (chapter Switch-on and reference run). The door is then once again ready for work.

The door is not designed for regular manual use.

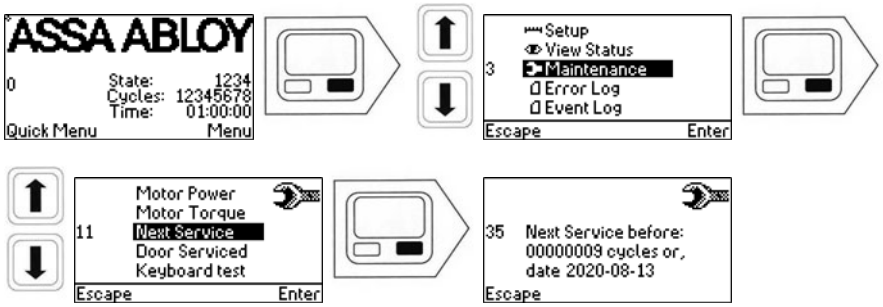
For doors with power on brake, the brake automatically releases if the main voltage is switched off, e.g. by switching off the main switch or disconnecting the CEE plug. You must initiate a reference search after the main voltage has been restored.
Initiate the reference search by pressing the "Open" button (chapter Switch-on and reference run). The door is then once again ready for work.

5.9 Service interval

If service is due, it is automatically shown with the following message. Every service interval indicated is recorded in the event log.

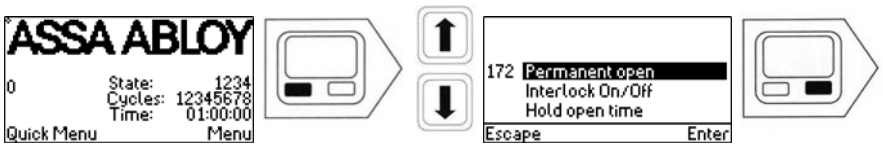


The date when the next service is due can be called up on the display.



5.10 Permanent opening of the door [1]

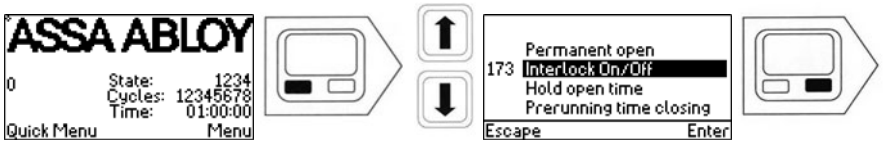
Activate permanent opening as following.

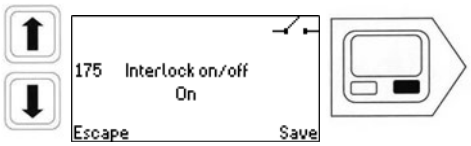


Deactivate permanent opening again by following the same sequence.

5.11 Deactivation of Interlock function [1]

Doors that are interlocked with one or more other doors have the option of deactivating the interlock function. This could be necessary if maintenance work is being carried out on one of the doors or the door shall be used as a single non-interlocked door, for example in summer.





Reactivate the interlock function by following the same sequence.

5.12 Crash-system

In order to minimize the damage of the door in case of impacts, doors can be equipped with a crash-system. If there has been an impact to the door and the crash sensor has been activated, the door is stopped immediately. The following error message appears:



Warning: Risk of personal injury and material damage!

Damaged parts of the door can endanger people or cause associated consequential damages.
After any impact to the door check thoroughly for possible damage.

5.12.1 Auto-repair crash system:

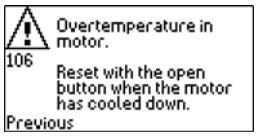
The door must be closed completely in hold to run mode. To do this, press the close button on the display until the crash mechanism is snapped-in in place again and the door stops autonomous in the lower end position. Another "Open" impulse will then make the door ready to use again.

5.12.2 Manually repairable crash system:

Snap-in the crash mechanism into the side frame manually. After this move the door to the lower end position to quit the error. The door is then ready for use again.

5.13 Thermal switch

To protect the drive motor three thermal switches are fitted in its windings. If the switches trip, the motor has become too hot and the drive is switched off immediately. The following message will then appear:



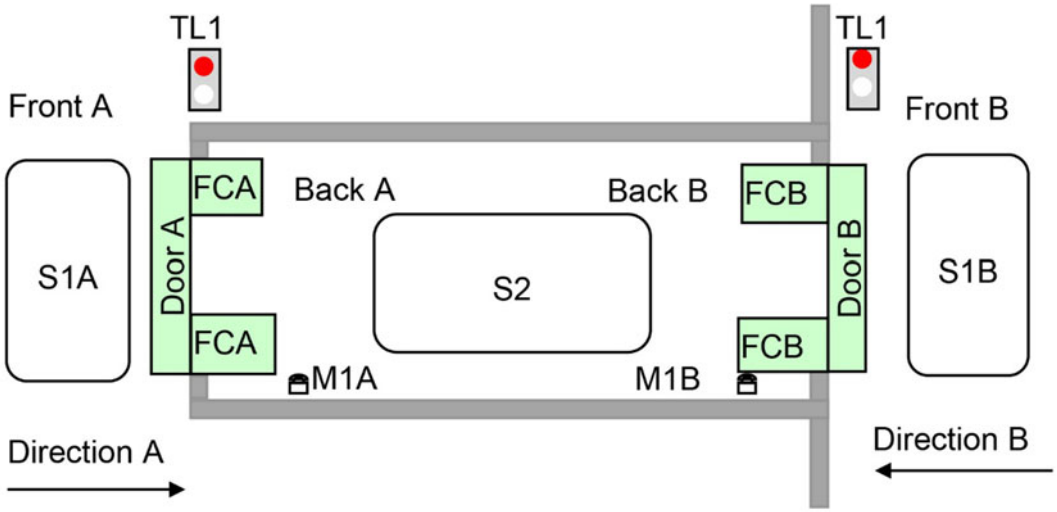
Once the motor has cooled down, acknowledge the error by pressing the "Open" button. Check the door and the control system for possible mechanical or electrical faults. Do not switch back on the control system until the motor has cooled down.

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5.14 Airlock function [1]

An Airlock consists always of a door A and door B and is controlled automatically by the airlock sequence. Each door has a separate control system. An airlock is used, e.g. to prevent draughts in external doors.

5.14.1 Airlock sequence [1]



Airlock sequence

#	**Description**
S1A	Actuator for Door A
S1B	Actuator for Door B
S2	Middle actuator in the airlock
FCA	Door line photocell Door A
FCB	Door line photocell Door B
M1A	Actuator for exiting the airlock Door A
M1B	Actuator for exiting the airlock Door B
TL1A	Traffic light for Door A
TL1B	Traffic light for Door B

5.14.2 Airlock passage from A to B with middle actuator

The airlock sequence can only be started if both doors are closed and the actuators are not activated. If the doors are equipped with traffic lights, both traffic lights must be red or off (depending on the parameter setting).
As soon as a vehicle activates actuator S1A, which is programmed with the function "Open with automatic closing after time (front)", Door A opens. After the door has been opened the airlock can be driven into. The door closes automatically after driving into the airlock and after the close after passage time has expired or after the hold open time, if the airlock has not been driven into.
Door B automatically opens if Door A is closed and middle actuator S2 of the airlock has been actuated. The middle actuator pulse is saved with the airlock function. This means that door B opens, even if the middle actuator is no longer activated, before Door A is closed. The airlock can now be exited through Door B which automatically closes after the hold open time or the close after passage time (depending on the actuator). If another vehicle drives through Door B into the airlock before Door B is closed, the airlock can be driven through in the other direction (but middle actuator (S2) must be activated once). Direction detection in the airlock and therefore the door initiated by the activation of the middle actuator is controlled by the door line photocell. If the door line photocell is actuated and then the middle actuator is activated, the direction is detected.

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5.14.3 Airlock passage from A to B without middle actuator

The airlock sequence can only be started if both doors are closed and the actuators are not activated. If the doors are equipped with traffic lights, both traffic lights must be red or off (depending on the parameter setting).
If a vehicle activates actuator S1A, which is programmed with the function "Open with automatic closing after time (front)", Door A opens. After the door has been opened the airlock can be driven into. The door closes automatically after driving into the airlock and after the close after passage time has expired or after the hold open time, if the airlock has not been driven into.
Door B opens automatically if Door A is closed, regardless of whether the airlock has been entered or not. The airlock can now be exited through Door B which automatically closes after the hold open time or the close after passage time (depending on the actuator). If another enters the airlock through Door B before Door B has started to close, direction detection is not set and Door A does not open. Direction detection in the airlock is activated by the actuator which opened the closed door or opens it while it is closing. Here it is possible that somebody enters the airlock and the 2nd door does not open automatically. Therefore, in this airlock option, a manual actuator for exiting the airlock should be provided at each door within the airlock (M1A and M1B).

5.14.4 Long vehicle function at doors with middle actuator



This function is required if the vehicle to pass through the airlock is longer than the distance between the two airlock doors.

The first door opens and the long vehicle can drive into the airlock. If the door line photocell of the opened door and the middle actuator (S2) are simultaneously activated, the second door opens automatically and the airlock can be completely driven through. The first door closes after the close after passage time and the second door after the hold open or close after passage time (depending on the actuator). This function is also possible from the opposite side.

5.14.5 Manual long vehicle function

There is a possibility to have a separate long vehicle function, like a push button or pull rope, connected to door A, independent of a middle actuator is used or not. See Programming the inputs (chapter Programming the inputs [1]). This long vehicle function is independent of the middle actuator and opens both doors without delay.

5.14.6 Permanent opening function

In several cases, e.g. during hot weather, it can be advantageous to use Door A only or Door B only. In this case the other door (here Door A) is set to permanent opening using the permanent opening function at the display (chapter Permanent opening of the door [1]) or, if installed, using the external permanent opening switch.
The opening pulse from the middle actuator (S2) then serves as a normal opening pulse for the door without permanent opening (Door B). The external actuator (here S1B) opens the door accordingly from the other side. The external actuator (here S1A) which is connected directly to Door A operated in permanent opening, does not have any function. In airlocks without middle actuator the external actuators (S1A or S1B) acts as opening pulse for the door without permanent opening.

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5.14.7 Direction detecting red traffic light on the outside of the airlock door

The red traffic lights are connected depending on the function "Traffic light red 24 V DC front" (24 V DC traffic light) or "Traffic light (red/green) front" (traffic light via relay contacts). The traffic lights via relay contacts require an additional MCC 2 output to switch off the traffic light if the door is closed.
Traffic light function settings: "Off if door is closed" or "Permanently on"
Airlock function from A to B:

Position of Door A	**Traffic light function of Door A**	**Position of Door B**	**Traffic light function of Door B**
Door A closed	Red = Off/On (*1) Green = Off	Door B closed	Red = Off/On (*1) Green = Off
Door A when opening	Red = On Green = Off	Door B closed	Red = On Green = Off
Door A open	Red = Off Green = On	Door B closed	Red = On Green = Off
Door A when closing	Red = On Green = Off	Door B closed	Red = On Green = Off
Door A closed	Red = On Green = Off	Door B when opening	Red = On Green = Off
Door A closed	Red = On Green = Off	Door B open	Red = On Green = Off
Door A closed	Red = On Green = Off	Door B when closing	Red = On Green = Off
Door A closed	Red = Off/On (*1) Green = Off	Door B closed	Red = Off/On (*1) Green = Off

(*1) = Depending on the setting selected for the traffic light.

5.14.8 Actuators and time setting

The minimum close after passage time, which is possible for an airlock door, depends on whether actuators S1A and S1B are detecting direction or not. Direction detecting actuators are, e.g. radar detectors and loops detectors. Non-direction detecting actuators are, e.g. infrared motion detectors and photocells.
After the airlock has been passed through (from A to B), actuator S1B sends a renewed opening pulse to door B if the close after passage time is set shorter than the time required to leave the actuator sensing range of S1B. This opening pulse starts the close after passage and hold open time again and in airlocks without middle actuator, Door A opens again after Door B is closed, if direction detection has been set with the opening pulse. Direction detecting actuators should be used to prevent this. Alternatively, the close after passage time can be set for so long that the object has the possibility of leaving actuator sensing range before the close after passage time has expired.
Due to saving of the middle actuator pulse even, e.g. a simple photocell can be used as the middle actuator. Direction detection is then set by the door line photocell as the middle actuator is not activated simultaneously.

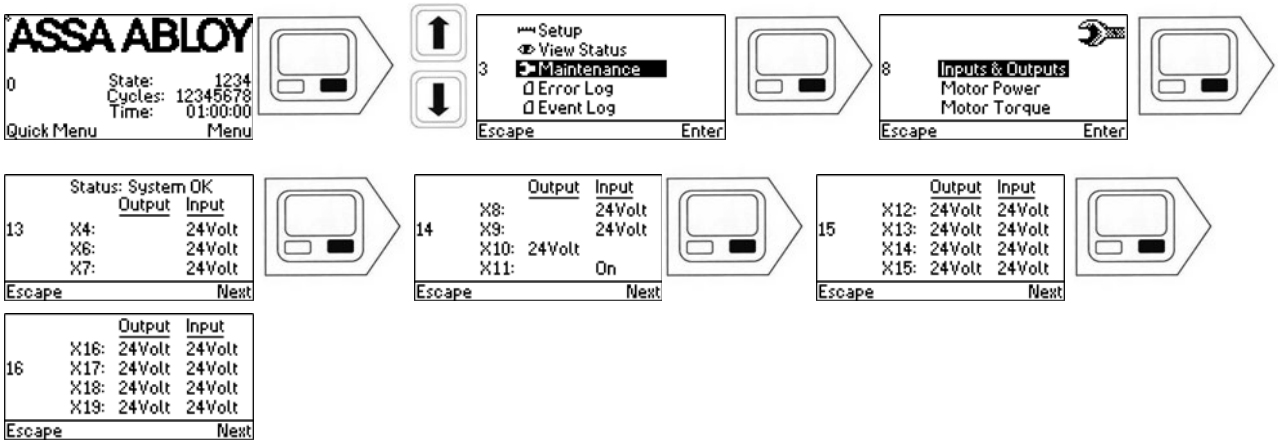
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6. Fault correction

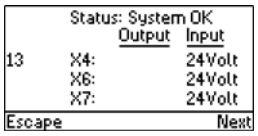
Contact the supplier or manufacturer for fault correction and repairs.
ASSA ABLOY Entrance Systems
Internet:www.assaabloyentrance.com
E-mail:info.aaes@assaabloy.com

6.1 Checking the inputs and outputs

On these display views it is easy to see what voltage is present at each input and output.



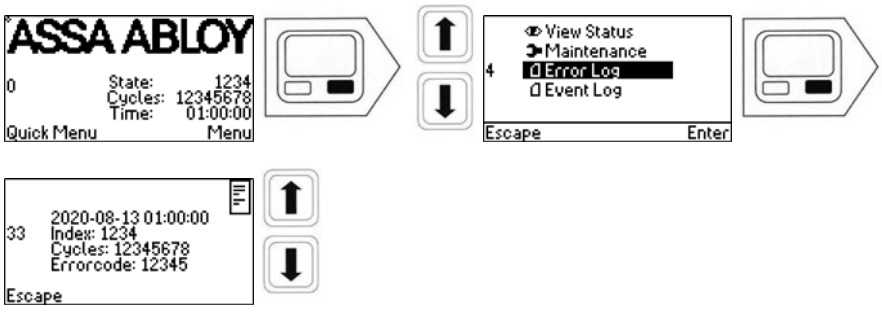
Example: In the electrical drawing the thermal switch is connected to X4. The following view appears on the display:



In this case the system status is OK, there is 24 V at the input of the control system and the thermal contact appears to be in order.

6.2 Error log readout

Information and error messages are recorded in the error log of the control system with the date and time.
The error log of the MCC 2 can be read-out as follows:



Every individual display view shows the date and time at which the error occurred, as well as the index, cycle status and the error code. The index is a running counter, which increases by one with each new error. The last occurred error is shown first. Use the two arrow keys on the display to scroll through all error messages.

Whenever the main switch is switched off a low voltage warning will be recorded in the error log!

The error log is divided into 2 arrays.
0 – 500 arrays of information (e.g. configuration changed)
500 - … error messages (e.g. emergency manual handle actuated)

Error code	Description of error
0	Text entry
1	Error log erased
2	System start
3	Programming PC connected
4	Programming PC disconnected
5	Setup of end positions started
10	Invalid data in the EEPROM
16	Door run 1 cycle (not activated)
17	Door run 10 cycles (not activated)
18	Door run 100 cycles (not activated)
19	Door run 1000 cycles (factory setting)
20	Reference limit switch position tolerance >10 impulses
21	Reference limit switch position tolerance >20 impulses
22	Reference limit switch position tolerance >50 impulses
23	Reference limit switch position tolerance >100 impulses
24	Reference limit switch position tolerance >150 impulses
30	IGBT temperature > 60°C (140°F)
31	IGBT temperature > 70°C (158°F)
32	IGBT temperature > 80°C (176°F)
33	IGBT temperature > 90°C (194°F)
34	IGBT temperature > 100°C (212°F)
41	Endurance test started
42	Endurance test stopped
43	State changed externally
44	Invalid positions
100	DSP, electrical safety edge activation
101	DSP, 18 V absent
103	DSP overvoltage warning
104	Overvoltage at DSP
105	DSP over temperature warning IGBT
106	Normal function (no reference search necessary)
107	Reference search after switch-on
108	Set up end positions
109	Overload warning
110	Safety function activated for closing impulse

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Error code	Description of error
111	DSP error
112	Valid door position saved
113	Door reversed by an opening function (Up) during closing
114	Door reversed by an opening function (UpDown) during closing
115	Door reversed by an opening function (UpButton) during closing
116	Door reversed by an opening function (UpAuto) during closing
117	Door reversed by an opening function (UpReduced) during closing
118	Door reversed by an opening function (UpAutoReduced) during closing
119	Door reversed by the main safety function (X11) during closing
120	Door reversed by the doorline photocell (X9) during closing
121	Door reversed by an additional safety device during closing
122	The main safety function (X11) active at manual close initiation
123	The doorline photocell (X9) active at manual close initiation
124	The additional safety active at manual close initiation
500	Invalid config file
503	Software logic error
504	DSP communication error
505	Software routines disturbed
506	Date or time out of valid range
507	Date/time out of valid range revised
508	Door moved from closed position without motor power
509	RAM test error
510	ROM test error
511	Closing speed higher than allowed
512	Communication timeout with the Anybus CompactCom module
513	Hand crank/manual lever input test error
514	Overtemperature in motor input test error
520-526	Communication error DSP – M16
1004	Electrical safety edge / pre-running safety photocell circuit interrupted during opening
1005	Electrical safety edge or the pre-running safety photocell activated
1006	Thermal contact tripped
1009	Movement of the door was not recognized
1010	Encoder registers unauthorised movement
1011	Emergency manual lever operated
1012	Broken wire/circuit to Reference limit switch (1012)
1013	Reference limit switch activated out of valid range
1014	Door locked by other door (airlock)
1015	Crash sensor has tripped

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Error code	Description of error
1016	Pass door open
1018	Safety 1 defect
1019	Door line photocell defect
1022	Safety 2 defect
1023	Internal frequency converter error
1024	Electrical safety edge / pre-running safety photocell interrupted in door open position
1025	Internal error
1026	Reference limit switch not activated with opened door
1027	Error in safety edge input
1028	Error in emergency stop input
1029	Input voltage at emergency stop input too high
1030	Earth fault in the 24 V circuit (low voltage)
1031	Earth fault in the 24 V circuit (high voltage)
1032	Opening height not reached during cyclical self-test (only for RR 300 Plus F+R)
1033	Breakaway function activated (only for US doors)
1100	Undervoltage – only in UPS mode activated
1101	Phase error
1102	Overvoltage
1104	Incorrect main voltage
1105	Overcurrent >25A
1106	Motor blocked or encoder without signal
1108	Over temperature
1110	Wrong hardware
1112	Brake circuit overcurrent
1115	3 Phase error
1200	Electrical safety edge or pre-running safety photocell activated

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6.3 Event log read-out

Important service messages and carried out service works are recorded in the control system's event log.
The event log of the MCC 2 can be read-out as follows:



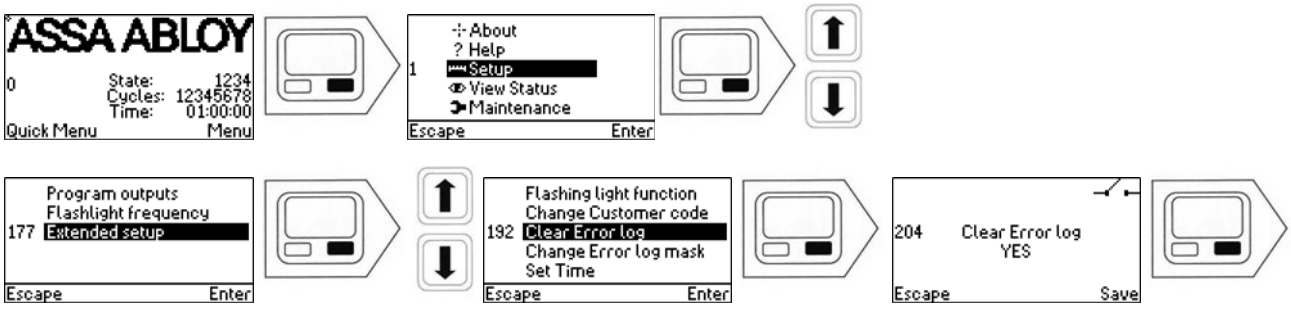
Every individual display view shows the date and time at which the event occurred, as well as the index, cycle status and the event code. The index is a running counter, which increases by one with each new event. The last occurred event is shown first. The two arrow keys on the display can be used to scroll through all event messages.

The following actions are recorded in the control system's event log:

Event code	Event
1	Event log cleared
2	Config file changed
3	Application software changed
5	Information note in event log added or changed
10	Door service required
11	Door serviced
20	Control system cycle counter changed
21	Door cycle counter changed
22	Function of inputs/outputs changed
23	Time changed

6.4 Clear error log [3]

The error log should only be cleared by ASSA ABLOY Entrance Systems Service as all existing error messages are permanently erased.



6.5 Altering the error mask [3]

Only for ASSA ABLOY Entrance Systems Service Team.

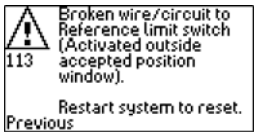
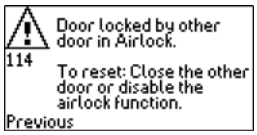
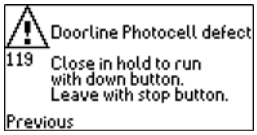
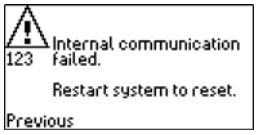
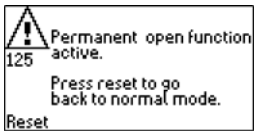
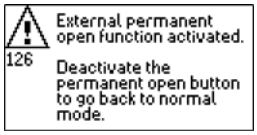
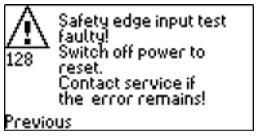
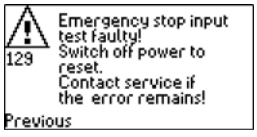
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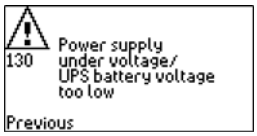
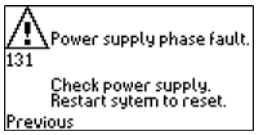
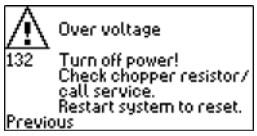
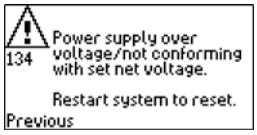
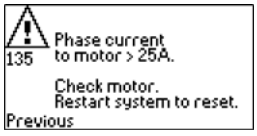
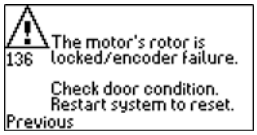
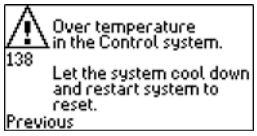
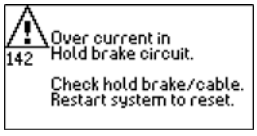
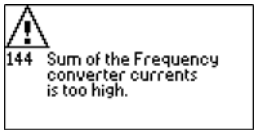
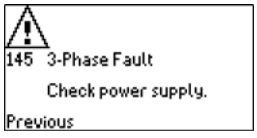
6.6 Error messages (only for doors with displays)

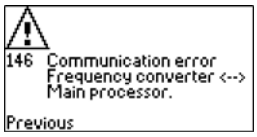
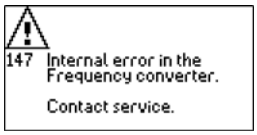
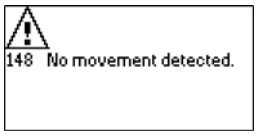
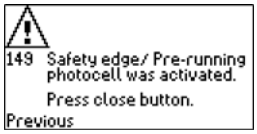
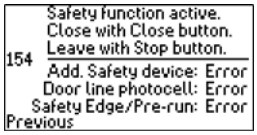
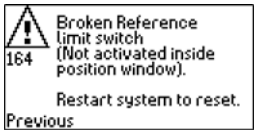
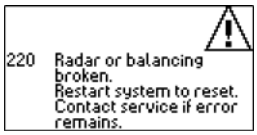
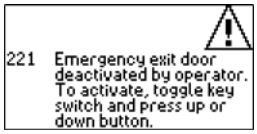
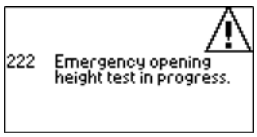
The following error messages are listed in ascending numerical order according to the display screen number. This number can be taken from the left-hand side of the display for every display message, regardless of the language. For example, 103 for "Door stopped".

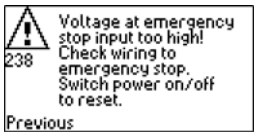
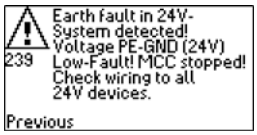
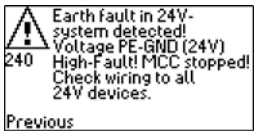
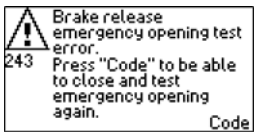
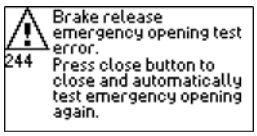
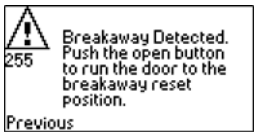
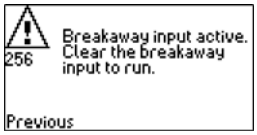
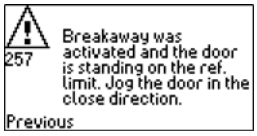
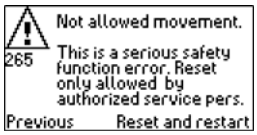
If no corrective action is specified, the error cannot be repaired or the error message appears again, please contact ASSA ABLOY Entrance Systems Service.

View-No	Display view	Corrective action
98	98 Open- or close-function active during Power up. Please remove or disconnect the signal(s). Escape	Check impulse actuator.
103	103 Door stopped. Press open or close. STOP Previous	Release stop button if necessary. Acknowledge by actuating open or close command.
104	104 Door stopped by panel. Press panel open or close. STOP Previous	Acknowledge by actuating open or close button on the display.
105	105 Safety edge/ Pre-running photocell activated Previous	Check pre-running safety photocell or electrical safety edge. Clean pre-running safety photocell. Acknowledge by actuating close button on the display.
106	106 Overtemperature in motor. Reset with the open button when the motor has cooled down. Previous	Switch off the control system, allow the motor to cool and, if necessary, clean the motor's ventilation slots.
107	107 Safety check failure Previous	Internal safety test faulty. RAM/ROM tests, speed monitoring or other internal safety tests faulty.
109	109 No movement detected. Restart system to reset. Previous	Encoder gives incorrect signals. The motor or the top roll is blocked.
110	110 Door interlocked by another door. To reset: Close the other door or disable interlock function. Previous	The door is locked by another door. Close the other door to reset this.
111	111 Hand crank attached or Hold brake manually released To reset: Remove crank or reset release handle Previous	Return the emergency manual lever to its original position or remove the emergency hand crank.
112	112 Broken wire/circuit to Reference limit switch Previous	Check wiring reference limit switch.

View-No	Display view	Corrective action
113		The reference limit switch was activated outside accepted door position window which is given by the encoder or the reference limit switch circuit is broken.
114		The door has been locked in place by the other air-lock door. To reset, close the other door completely or deactivate the air-lock function.
115		For doors without an auto-repair crash system close the door completely after repairs to acknowledge the error message. For doors with an auto-repair crash system close the door until the bottom profile engages automatically again and the lower end position is reached.
119		Exclude influence from other infrared (IR) sensors and check alignment of the door line photocell.
122		Check function and wiring of additional safety devices.
123		Please call the ASSA ABLOY Entrance Systems Service Team.
124		Check function and wiring of electrical safety edge / the pre-running safety photocell.
125		Permanent opening has been activated. Press left function key for reset.
126		External permanent opening has been activated. To deactivate this, remove the permanent opening signal.
127		The encoder detects an unauthorized movement of the door.
128		Please call the ASSA ABLOY Entrance Systems Service Team.
129		Please call the ASSA ABLOY Entrance Systems Service Team.

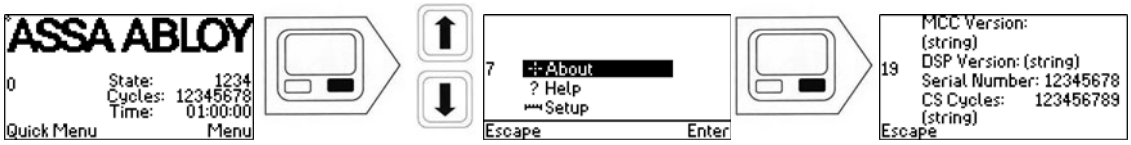
View-No	Display view	Corrective action
130		Undervoltage detected in the main voltage supply. Check the main voltage. Verify that the original config settings for the door are correct. Recharge the batteries in case of doors with UPS.
131		Check main voltage and supply stability. To reset switch the main switch off and back on again.
132		Overvoltage in the control system. Turn off the main switch and check the brake resistor, the main voltage and control settings of main voltage.
134		Check the supply voltage and the parameter for the main voltage.
135		The motor current is too high. Check that the door mechanism is running smoothly.
136		Check that the motor and the door mechanism is running smoothly. Check alignment and function of the encoder.
137		The control system is overloaded. Check that the door mechanism is running smoothly.
138		Overheating of the control system. Switch off the main switch and allow the control system to cool.
140		Please call the ASSA ABLOY Entrance Systems Service Team.
142		Check the brake and the brake cable.
144		Check that the door mechanism is running smoothly.
145		3-phase voltage failure or break-down of the supply voltage occurred. Check the supply voltage and the fuses.

View-No	Display view	Corrective action
146		Please call the ASSA ABLOY Entrance Systems Service Team.
147		Please call the ASSA ABLOY Entrance Systems Service Team.
148		The door is blocked or the position encoder is sending incorrect signals. Check that the door mechanism is running smoothly. Check the function of the encoder.
149		Check electrical safety edge / the pre-running safety photocell. Clean pre-running safety photocell. Press the "Open" or "Close" buttons to reset.
151		Release the emergency stop or remove emergency hand crank and start the door using the "Open" or "Close" command.
153		Acknowledge by actuating open command.
154		The door's safety function is active. This means that there is something wrong with one or more of the door's safety devices. The current status of the safety devices connected can be seen on the display. Once all safety devices are functioning correctly again exit or leave the mode by pressing the stop button.
164		Check reference limit switch and adjust if necessary.
220		Only for emergency exit door ASSA ABLOY RR 300 plus F+R Check springs, belts and spring/belt break switches. Check wiring of the radar and spring/belt break switches.
221		Only for emergency exit door ASSA ABLOY RR 300 plus F+R Toggle deactivation key-switch to switch-on emergency exit function. If message continues to be displayed, check the switch and wiring and replace if necessary.
222		Only for emergency exit door ASSA ABLOY RR 300 plus F+R No measures are necessary after successful completion. If performance of the self-opening-test is faulty, consider other fault messages.

View-No	Display view	Corrective action
238		The voltage at the emergency stop input is too high. Check wiring of emergency stop circuit.
239		An earth fault has been detected in the 24 V circuit. Check the wiring of all 24 V loads.
240		An earth fault has been detected in the 24 V circuit. Check the wiring of all 24 V loads.
243		Only for emergency exit door ASSA ABLOY RR 300 plus F+R Door can be released with right function key "Code" and input of the customer code. The self-tests are then repeated. The message 222 is indicated afterwards.
244		Only for emergency exit door ASSA ABLOY RR 300 plus F+R Press Close button for renewed checking of the self-opening height. No measures are necessary after successful completion.
255		Only for US doors: There has been a door impact. Acknowledge by moving door to upper end position of the door (breakaway position).
256		Check breakaway-sensors.
257		Follow instructions in the display view. Check the function and wiring of the reference limit switch.
260		Replace the batteries of the radio system only with equivalent batteries. Operation of the door in hold to run mode is possible.
265		Please call the ASSA ABLOY Entrance Systems Service Team.

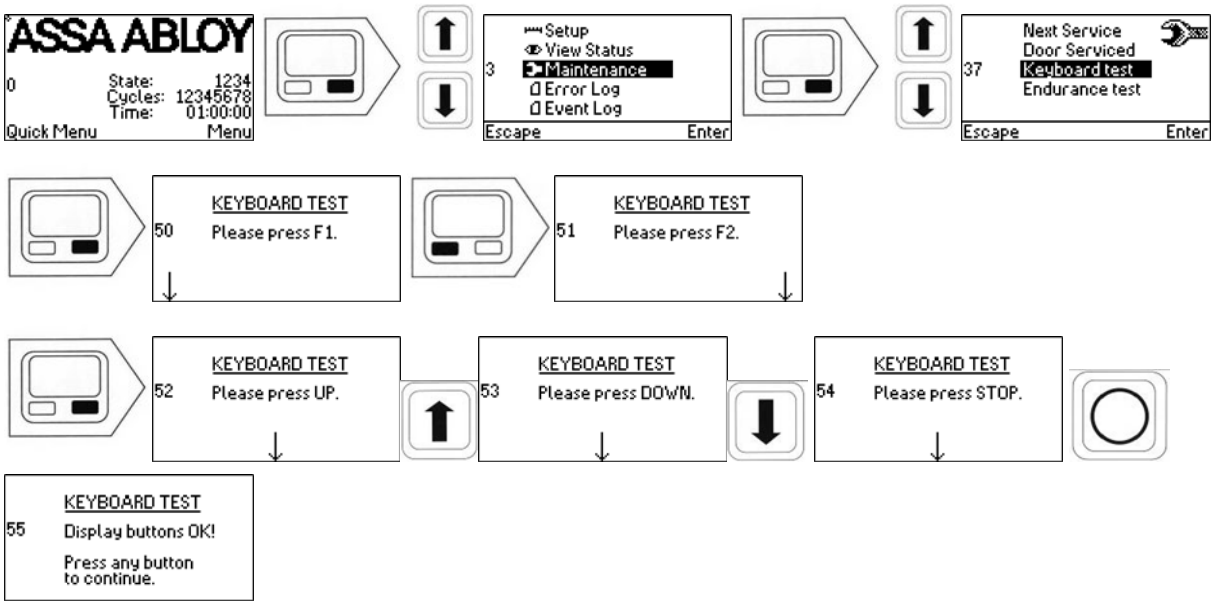
6.7 Hardware and software version readout

To read-out the hardware and software versions use the following sequence:



6.8 Keyboard test

Use this test to check whether the individual keys on the display are working.



6.9 Technical support



The display lists the contact information for technical support or the website on which the latest contact information for local technical support can be retrieved.

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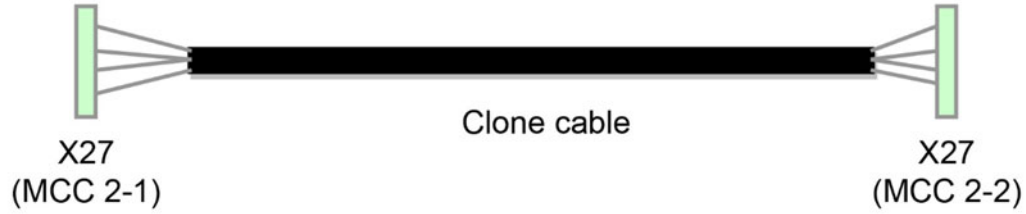
6.10 Clone function when replacing the control system

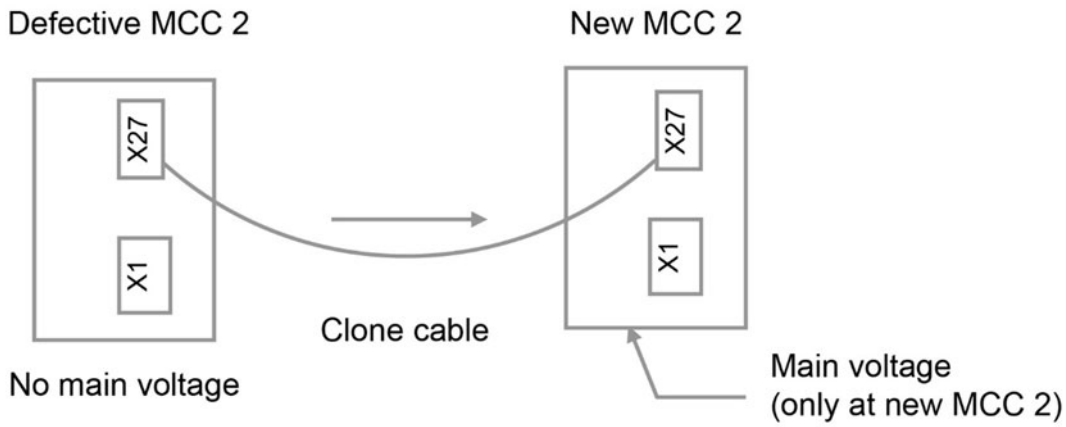
You can copy the software from one MCC 2 control to another MCC 2 with the clone function.

The clone function from an MCC to an MCC 2 or vice versa is not possible.

To clone a faulty MCC 2 system the following components are needed:

- A new MCC 2 control system with at least pre-installed boot software.
- The clone cable (therefore contact ASSA ABLOY Entrance Systems Service).
- An MCC 2 display with cabling (if not already installed at the door)





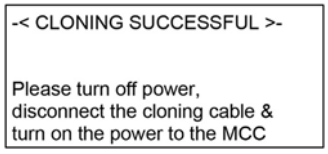
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Follow the step-by-step instruction to perform a complete cloning:



The only cable that shall be connected to the faulty MCC 2 is the cloning cable. The defective MCC 2 is not connected to the net voltage (main voltage)

1) Turn off all power.
2) Disconnect and remove the broken MCC 2 from the door.
3) Install the new MCC 2 including the complete electrical-connections.
4) Connect the cloning cable to both, the new MCC 2 and to the faulty MCC 2 (plug X27).
5) If no MCC 2 display is normally connected to the door then temporary connect an extra one to the new MCC 2.
6) Turn on the power at the installed MCC 2.
7) Select "CLONE" on the display.
   Wait until cloning is finished and following message is displayed.



8) If the cloning has not been performed successfully the procedure can be restarted from point 6 by cycling the power.
9) Turn off the power.
10) Remove the faulty system, the cloning cable and eventually the extra MCC 2 display.
11) Turn on the power.
12) Test the door.

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7. Technical data and control system label

7.1 General data

Housing material	anodized aluminium
Dimensions of housing WxHxD with end plates, but without cable glands	220 x 340 x 100 mm (8,7" x 13,4" x 4")
Rated input voltage	3~ PE; 50-60 Hz 208 VAC - 480 VAC Tolerance ±10%
Max. input current	8 A (FLA)
Rated output power	3 kVA
Operating temperature range	-20°C - +40°C -20°C - +45°C RP-doors
Protection class	IP 55
Internal fuse	3x10 A fast acting (6,3x32 mm) Internal fuses are allowed to be changed by ASSA ABLOY Service only according to chapter Electrical installation requirements!
External fuse by customer	10 - 16 Ampere B or C characteristic or comparable
Residual current protection (RCCB`s) (only if needed)	AC/DC sensitive short-time delayed type B Observe the instructions in chapter Electrical installation requirements!
Residual current caused by MCC 2	≤ 7mA according EN 60335-1
Control voltage / external supply	24 VDC / max. 500 mA (PELV) with display max. 450 mA
Standby consumption	10 W
Brake voltage	96 - 196 VDC / max. 300 mA
Programmable inputs	8
Programmable outputs	8
Weight (with endplates)	approx. 4,7 kg
Conformity Europe	2006/42/EC, 2014/30/EU 2014/35/EU, 2011/65/EU EN 13241-1:2003+A2:2016 EN ISO 13849-1:2016, EN 12978:2009, EN 12453:2017, EN 60335-2-103:2015 Confirmed by external, independent testing institute
Motor cable length	Max. 3 m
Installation class	Industrial environment (according DIN EN 61000-6-4) For residential, commercial and light-industrial environments (according DIN EN 61000-6-3) additional PFC filter is necessary to fulfill the EMC requirements (see chapter 8.4).

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7.2 Classifications of safety circuits to EN ISO 13849-1

Note that the MTTFd below is for the MCC2 unit only. External sensors, cabling etc. are not included.

Function	Category	MTTFD	DCAVG	Performance Level
Input emergency stop (X72)	3	>100 years	medium	d
Input OSSD emergency stop (X73)	3	>100 years	medium	d
Input electrical safety edge (X11) or the pre-running safety photocell (X11/12)	2	>100 years	medium	d
Input doorline photocell (X9/10)	2	>100 years	medium	d



Warning: Risk of injury!

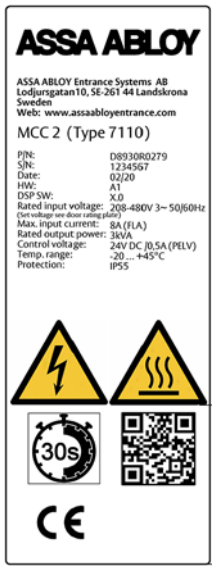
To fulfil the safety requirement for one additional safety on one of the 8 general input (X12-19) it is essential to comply with both following conditions.
a.) The test output for additional safety (additional safety test voltage) needs to be activated and
b.) Not more than one of the 8 inputs are used as additional safety. The use of furthermore inputs is not permitted in this case.
For safety reasons, this function can only be accessed by our specially trained safety experts.

Function	Category	MTTFD	DCAVG	Performance Level
Additional Safety Device (X12-19)	2	>100 years	medium	d (note warning before!)

Calculated MTTFd more than 100 Years has been limited to 100 years according to EN ISO 13849-1 4.5.2.

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7.3 Labeling, door and control system type plate (example)



Exemplary control system type plate

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8. Optional MCC 2 – Units

8.1 Interface board

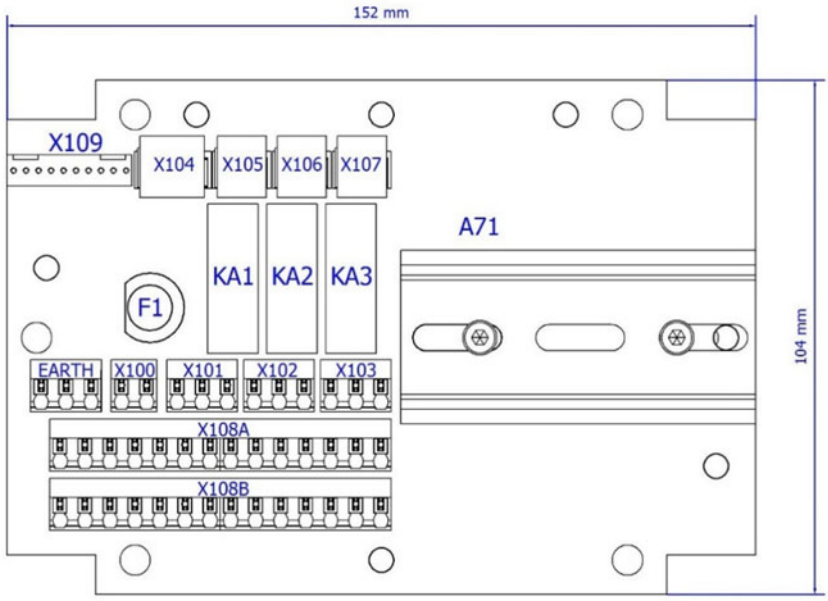
The MCC 2 can be supplemented with a multifunctional interface board which increases the scope of the control system. There are 3 dry relay changeover contacts (8A, 250V~ / 10A, 240V-) on the interface board for controlling external components, which are controlled directly through the MCC 2 outputs. In addition, the board offers the possibility to move 3 inputs from the MCC 2 onto the interface board. This enables add-ons, e.g. loop detector or radio receiver to be mounted on the top-hat rail and to be directly wired with the inputs of the MCC 2. The terminal block X108 (terminal 1-14) can be used e.g. as an interconnection terminal block.

8.1.1 Terminal assignment of the interface board

Danger: electrical shock!
Infringement of protection class
If used, connect the protective earth terminal strip (Earth) to the existing protective earth connection.

Cross-section, spring-loaded terminals X100-X103, X108, protective earth connection:
0.14-1.5 mm^2 without wire end sleeve,
0.14-1.5 mm^2 with wire end sleeve without plastic collar to DIN 46228-1
0.14-0.75 mm^2 with wire end sleeve with plastic collar to DIN 46228-4

Cross-section, spring-loaded terminals X104-107:
0.14-1.5 mm^2 without wire end sleeve,
0.14-1.0 mm^2 with wire end sleeve without/with plastic collar to DIN 46228-1/-4



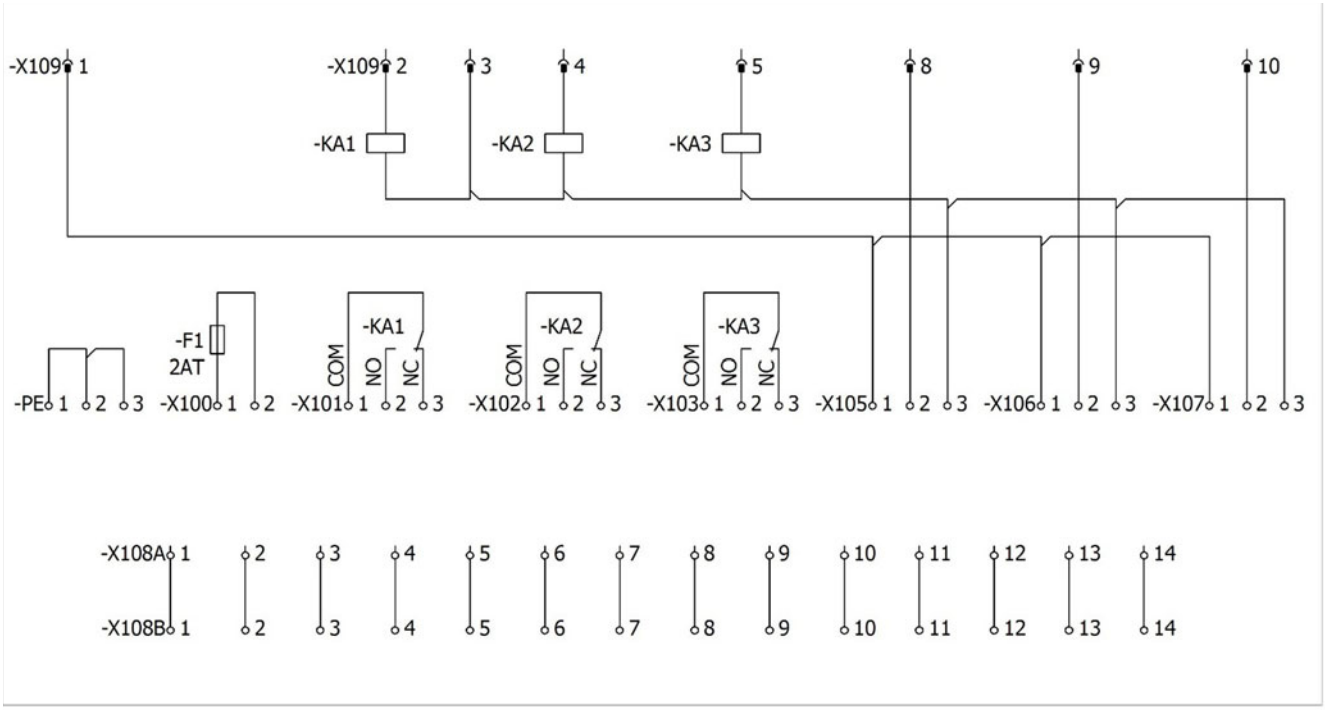
Interface board

Terminal	Description	Pin assignment
X 100	Connection of fuse F1	T2A => Terminal 1 and 2
X 101	Dry changeover contact of relay KA1	1 = COM, 2 = NO, 3 = NC
X 102	Dry changeover contact of relay KA2	1 = COM, 2 = NO, 3 = NC

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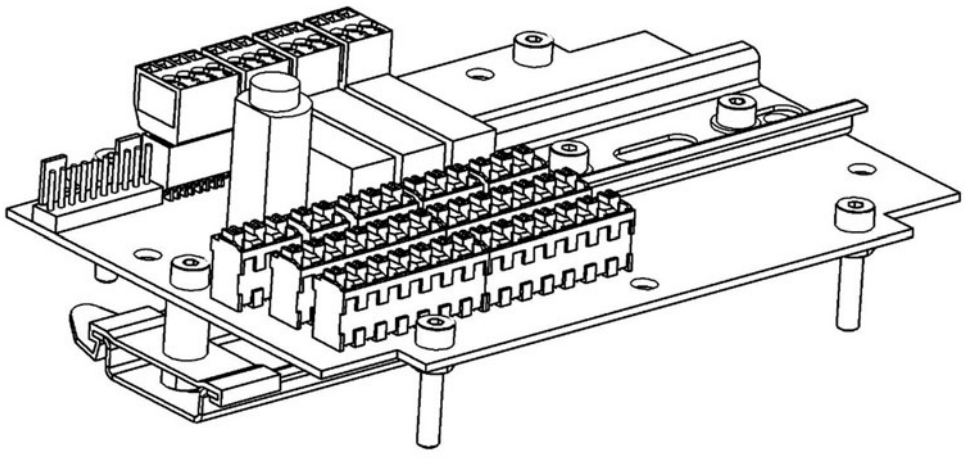
Terminal	Description	Pin assignment
X 103	Dry changeover contact of relay KA3	1 = COM, 2 = NO, 3 = NC
X 104	(not used)	
X105-X107	Programmable inputs of the MCC 2	1 = 24V DC, 2 = Input, 3 = GND, see order-related electrical drawing
X 108	interconnection terminal block with 14 terminals	
X 109	Connecting cable to the MCC 2	
PE/Earth	Terminal for protective earth connection	

8.1.2 Schematic electrical drawing of the interface board



8.1.3 Interface board for top-hat rail mounting

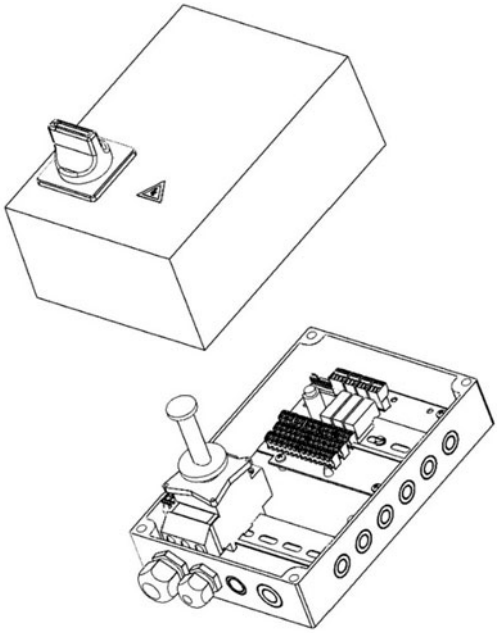
This variant of the interface board is designed to snap on an external top-hat rail in an existing housing.

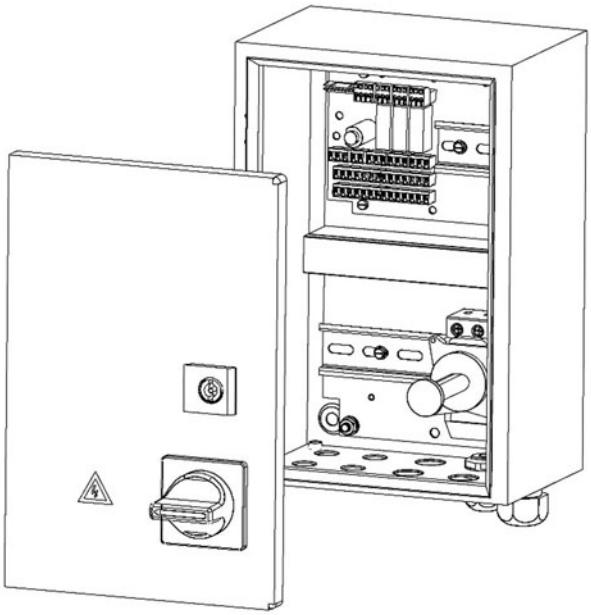


Interface board for top-hat rail mounting

8.1.4 Interface board in combi box

The interface board is mounted in a plastic or metal combi box housing.





Plastic combi box versionMetal combi box version

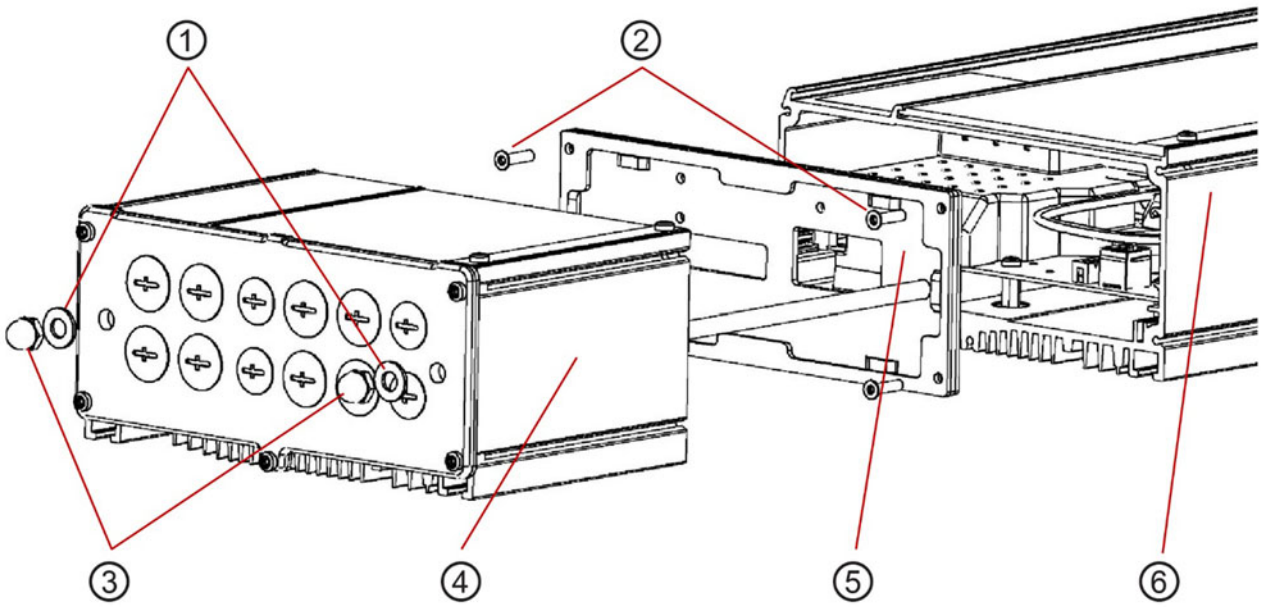
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8.2 MCC 2 extension

The MCC 2 can be supplemented with a multifunctional extension housing which increases the scope of the control system and provides reserve space for additional components.

8.2.1 Mechanical installation of the MCC 2 extension

As shown, screw the prepared connection plate onto the MCC 2 using the four countersunk screws. After that push the extension housing onto the connection plate and fix it using the two cap nuts.



1) Plastic plain washers
2) Countersunk screws
3) Cap nuts
4) MCC 2 extension housing
5) Connection plate
6) MCC 2

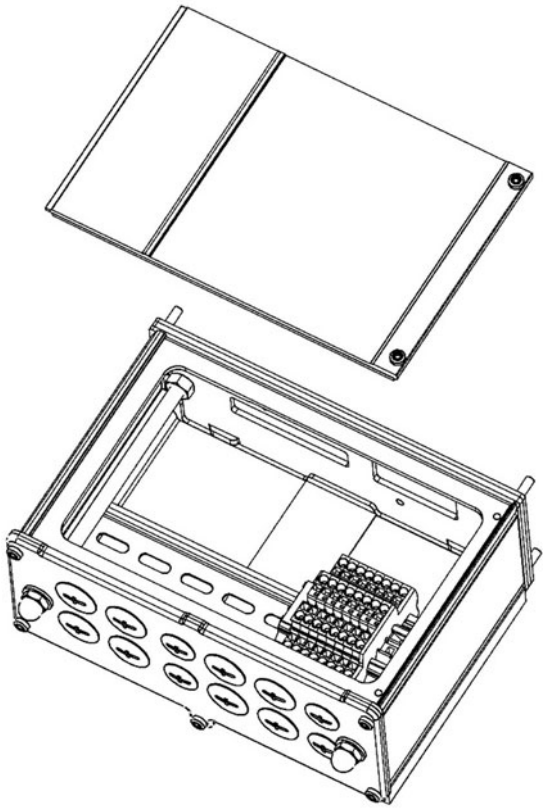
Notice: Risk of material damage!

Destruction of the MCC 2 and the extension board by moisture.
To seal the housing, ensure that the two plastic plain washers are used under the two cap nuts.

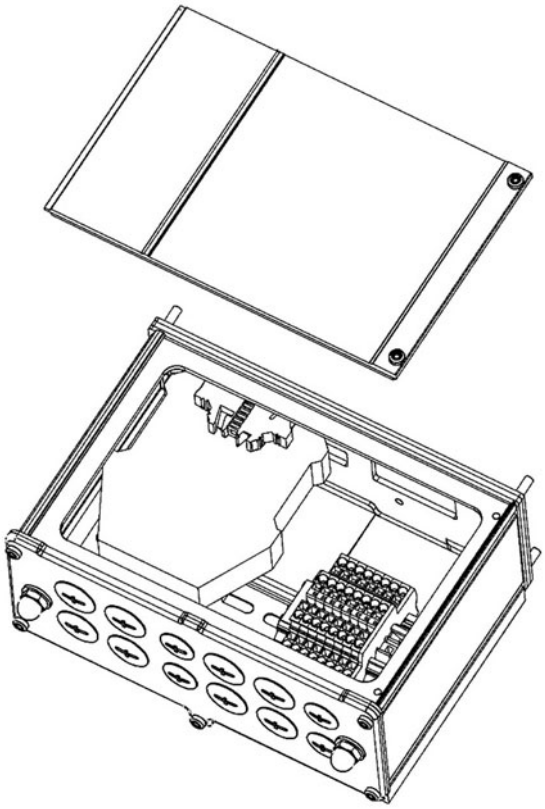
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8.2.2 MCC 2 extension with terminal block

For the connection of the external safety relay for the door safety switches an extra terminal block can be built in the extension housing (details see electrical drawing).



Extension with terminal block



Extension with terminal block and safety relay

8.2.3 MCC 2 extension with extension card

The MCC 2 extension card is mounted directly in the extension housing. It enables various plug-on boards for radio receivers and loop detectors to be directly installed. The connection with the MCC 2 is made using a 40-pin ribbon cable and therefore reduces the wiring required for the installation. With the MCC 2 extension it is possible to connect the whole sensor/actuator wiring at the door to the extension and, at the customer's request, to guide the inlets for this through the extension housing.
There are 4 dry relay changeover contacts (6A, 250V~ / 3A, 30V-) on the extension board for controlling external components and which are controlled directly through the MCC 2 outputs.

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8.2.3.1 Terminal assignment of the MCC 2 extension board

Danger: Serious injuries at the unsecured door!

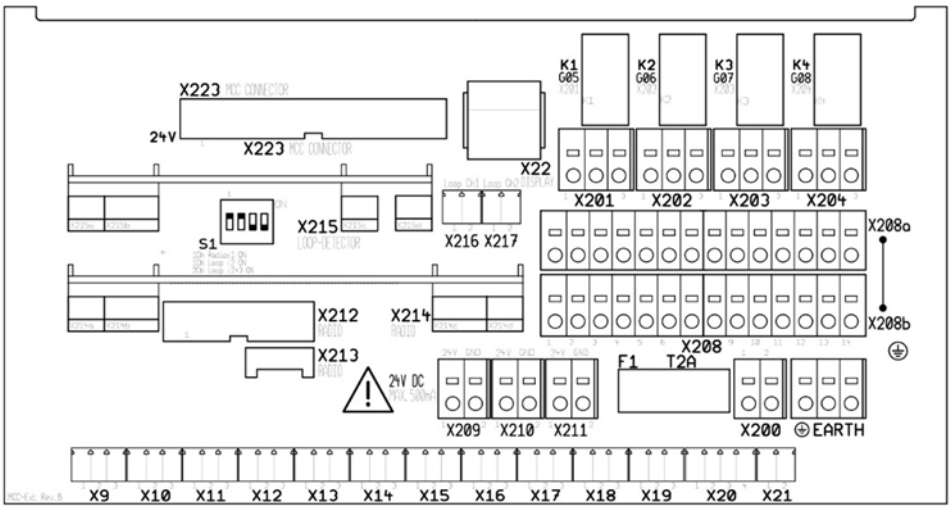
Loss of the CE conformity.
The inputs and outputs of the extension board and the MCC 2 are interconnected in parallel. A simultaneous use can lead to a loss in safety and to unexpected behaviour of the door.
The corresponding in- and outputs of the extension board and the MCC 2 should not be wired/used in parallel.
Observe electrical drawing of the door.
Simultaneous use according to order-related electrical drawing is allowed.

Danger: Risk of electrical shock!

Infringement of protection class!
If used, connect the protective earth terminal strip (earth) to the existing protective earth connection.

Cross-section, spring-loaded terminals X9-21, X216-217:
0.14-1.5 mm^2 without wire end sleeve,
0.14-1.0 mm^2 with wire end sleeve without/with plastic collar to DIN 46228-1/-4

Cross-section, spring-loaded terminals X200-211, protective earth connection:
0.14-2.5 mm^2 without wire end sleeve,
0.14-2.5 mm^2 with wire end sleeve without plastic collar to DIN 46228-1
0.14-1.5 mm^2 with wire end sleeve with plastic collar to DIN 46228-4



Extension card

Terminal	Description	Pin assignment
X 223	Plug-in connector for connection with the MCC 2	
X 9 – X 21	Cable connectors for sensors/actuators parallel to the plugs on the MCC 2	
X 22	Alternative connector for display	
X 200	Connection of the internal fuse F1 (250V T2A)	
X 201	Changeover contact relay K1 - MCC 2 output GO5/X16	1 = COM, 2 = NO, 3 = NC
X 202	Changeover contact relay K2 - MCC 2 output GO6/X17	1 = COM, 2 = NO, 3 = NC
X 203	Changeover contact relay K3 - MCC 2 output GO7/X18	1 = COM, 2 = NO, 3 = NC

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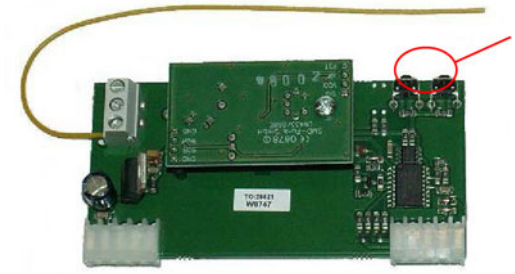
Terminal	Description	Pin assignment
X 204	Changeover contact relay K4 - MCC 2 output GO8/X19	1 = COM, 2 = NO, 3 = NC
X 208	interconnection terminal block with 14 terminals Terminals X208a and X208b are individually bridged internally!	
X209– X211	Supply terminals 24V DC for external loads	
X212 - X214	Slot for plug-in radio receiver	
X 215	Slot for loop detector 1 channel / 2 channel	
X 216	Terminal for loop 1 (twist loop wire up to the terminal!)	
X 217	Terminal for loop 2 (twist loop wire up to the terminal!)	
Earth	Terminal for protective earth connection	

8.2.3.2 Plug-on modules for MCC 2 extension

The MCC 2 extension enables various plug-on modules for radio receivers and loop detectors to be directly installed. All slots provided for this are specially designed for the described plug-on modules. Plugging in other modules can cause damage or destruction of the MCC 2 control system, the extension as well as the add-on modules. Only plug in or pull off the plug-on modules if the control system is switched off (disconnected from the power supply).
The respective plug-on modules can only be attached to the corresponding slot and only in one direction on the extension. If it is necessary to remove a plug-on module from the socket, this is made easier by alternately tilting the module.

8.2.3.3 Radio receiver 433 MHz

2 channel 433 MHz radio receiver, whereby only one channel is used on the MCC 2 extension.



Radio receiver module

Teaching-in of the radio receiver:
Press teach-in button of radio receiver and then actuate transmitter.
Up to 32 transmitter can be teached-in.
To delete the teached-in transmitter press teach-in button until LED lights up!



Observe manual of the radio receiver!

8.2.3.4 Loop detectors

Loop detector 1 channel / 2 channel as actuators for the detection of vehicles.



Loop detector module

Sensitivity:
By adjusting the sensitivity, you determine a change of inductivity for each channel, which a vehicle has to cause in order to set the appropriate output of the detector.
Sensitivity adjustment is done separately for each channel with the help of 2 DIP-switches.

Sensitivity level	**Channel 1: DIP-switch 1, 2; Channel 2: DIP-switch 5, 6**
1 (0.27 % Δf/f)	 **OFF/OFF**
2 (0.09 % Δf/f)	 **ON/OFF**
3 (0.03 % Δf/f)	 **OFF/ON**
4 (0.01 % Δf/f)	 **ON/ON**

Hold time:
Adjust the holding time with the help of DIP-switches 3 and 7. After the holding time has expired, a "loop free" signal is emitted, followed by an automatic rebalancing of the loops. The holding time starts as soon as the loop is seized.

Hold time	**Channel 1: DIP-switch 3, Channel 2: DIP-switch 7**
5 min	 **OFF**
infinite	 **ON**

Frequency adjustment and readjustment:
Adjust the actual frequency of the detector in two levels with the help of DIP- switches 4 and 8.

Frequency	**Channel 1: DIP-switch 4, Channel 2: DIP-switch 8**
low	 **OFF**
high	 **ON**

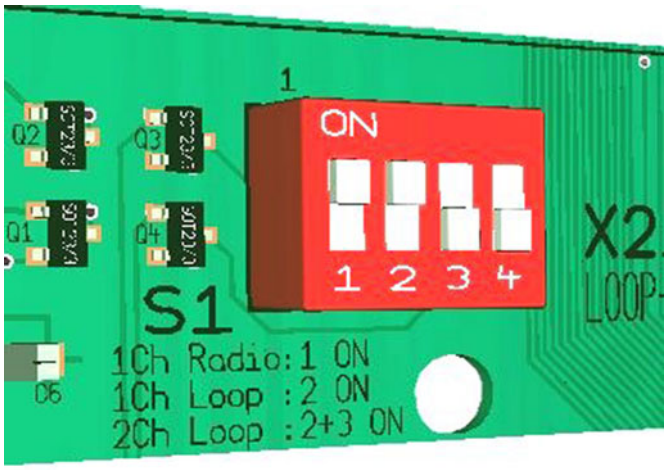
The admissible frequency range is 30 kHz to 130 kHz. The frequency depends on the loop geometry, number of turns, inductivity resulting from the loop supply line and the chosen frequency level.
A rebalancing can be manually triggered by changing the frequency adjustment of a channel.
As soon as switched on, the detector automatically carries out an adjustment of the loop frequency.
In case of a short-term voltage loss of <0,1 s no rebalancing will take place.



Observe the manual of the loop detectors!

8.2.3.5 Setting the slide switches for plug-on modules

These DIP-switches are positioned on the expansion card.



Slide switch

Warning: Risk of personal injury and material damage!

Disturbance of the general operation and unexpected behaviour of the door.
The inputs and outputs of the extension board and the MCC 2 are interconnected parallel. A simultaneous use of the inputs and outputs can lead to a disturbance of the general operation and unexpected behaviour of the door.
The appropriate inputs to the MCC 2 and the extension board should not be used for other impulse actuators when using the plug-on modules.
Observe electrical drawing of the door.
Double assignment according to order-related electrical drawing is allowed.

Radio receiver 1 channel
Switch slide switch 1 to ON position!

With the switching on of dip switch 1, input GI8 (X19) of the MCC 2 is in use and can be used for other impulse actuators in exceptional cases only.

Loop detector 1 channel
Switch slide switch 2 to ON position!

With the switching on of dip switch 2, input GI7 (X18) of the MCC 2 is assigned and can be used for other impulse actuators in exceptional cases only.

Loop detector 2 channel
Switch slide switches 2 and 3 to ON position!

With the switching on of dip switches 2 and 3, the inputs GI6 and GI7 (X17,X18) of the MCC 2 are assigned and can be used for other impulse actuators in exceptional cases only.

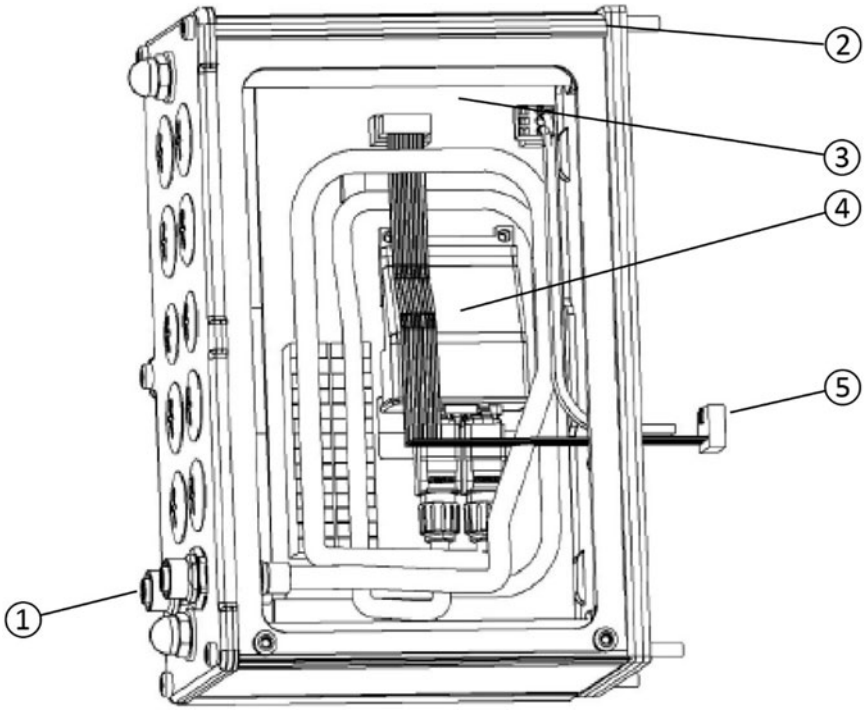
Slide switch 4 does not have any function!

8.3 MCC 2 extension with field bus module

The field bus module of the MCC 2 offers the possibility to connect the MCC 2 to different field bus systems. The corresponding fieldbus standard is set by selecting the used Anybus Compactcom (Anybus CC) communication module.

8.3.1 Structure of field bus module

The fieldbus module consists of a mainboard, an Anybus CC communication module and the MCC 2 extension housing, which is flanged at the MCC 2.
The communication of the field bus module with the MCC 2 control system is done by a 10-pole ribbon cable. The 24 V supply is from the MCC 2 and is to be connected to the plug X2 of the mainboard.
Depending on the field bus system the connection to the field bus is done by M12 female or male plugs which are installed in the flange-plate.

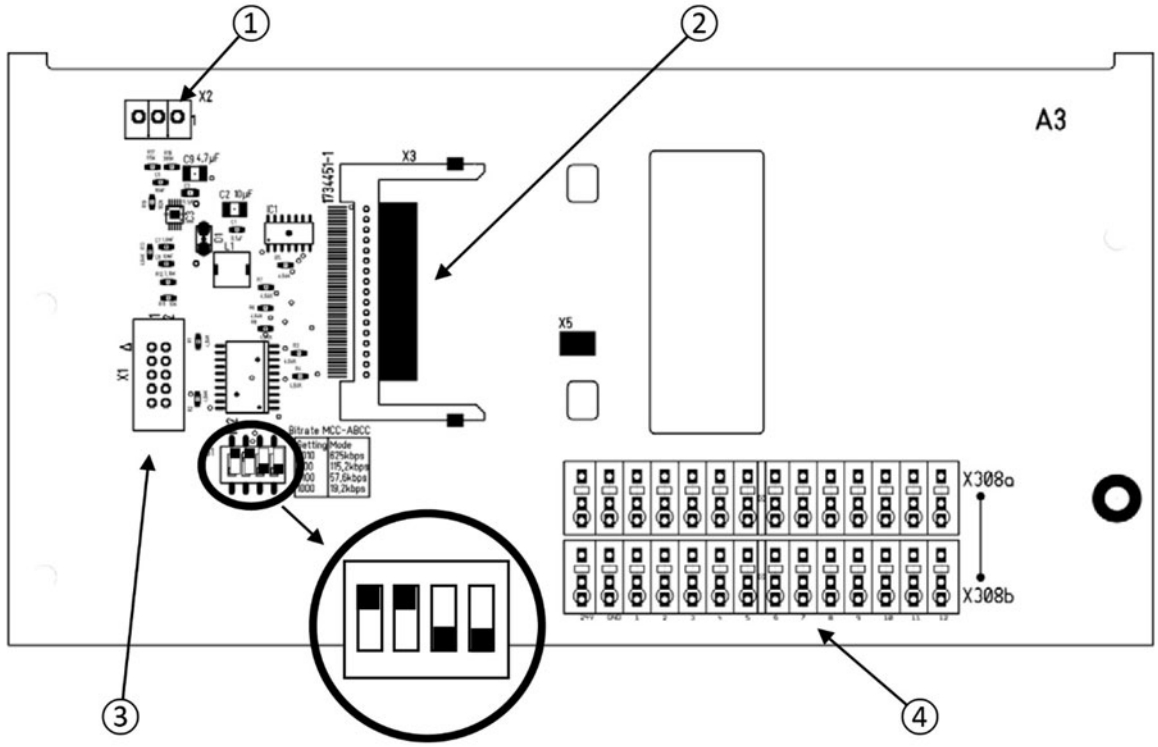


1) Field bus interconnection
2) MCC 2 extension housing
3) Mainboard
4) Anybus CC communication module
5) MCC 2 interconnection

Field bus module with PROFINET 2-Port

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8.3.1.1 Mainboard



Field bus module main board
1) Connector for supply voltage
2) Slot for Anybus CC module
3) Connector for MCC 2 interconnection
4) Terminal block for Safety relay

The DIP-switches on the mainboard for the transfer rate between MCC 2 and bus module shall be set to 115,2 kbps (1100).

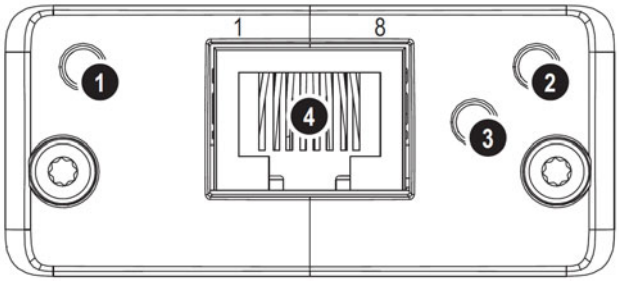
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8.3.1.2 PROFINET 1-Port (AB6215)

For the PROFINET 1-port bus structure an external network-switch is necessary.
For this solution the connection to the field bus is done via the flange plate with the integrated M12 female plugs (D-coded).

LED function PROFINET 1-Port (AB6215):
Front view:

No.	Description
1	Network Status LED
2	Module Status LED
3	Link/Activity LED
4	EtherNet Interface



Network Status LED

A test sequence is performed on this LED during startup.

LED State	Description	Comments
Off	Offline	• No power • No connection with IO controller
Green	Online (RUN)	• Connection with IO controller Established • IO Controller in RUN state
Green, flashing	Online (STOP)	• Connection with IO controller Established • IO Controller in STOP state

Module Status LED

A test sequence is performed on this LED during startup.

LED State	Description	Comments
Off	Not initialized	No power - or - Module in 'SETUP' or 'NW_INIT' state
Green	Normal operation	Module has shifted from the 'NW_INIT' state
Green, 1 flash	Diagnostic event(s)	Diagnostic event(s) present
Green, 1 Hz	Blink	Used by engineering tools to identify the node on the network
Red	Exception error	Module in state 'EXCEPTION'
Red, 1 flash	Configuration error	Expected Identification differs from Real Identification
Red, 2 flashes	IP address error	IP address not set
Red, 3 flashes	Station name error	Station Name not set

LED State	Description	Comments
Red, 4 flashes	Internal error	Module has encountered a major internal error

LINK/Activity LED

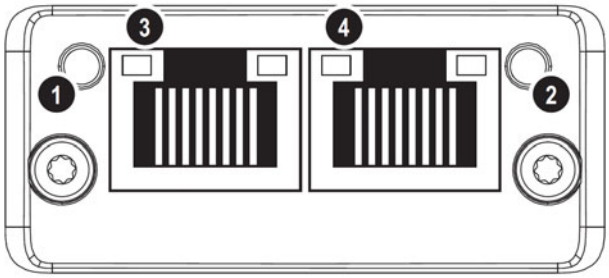
LED State	Description	Comments
Off	No Link	No link, no communication present
Green	Link	EtherNet link established, no communication present
Green, flickering	Activity	EtherNet link established, communication present

8.3.1.3 PROFINET 2-Port (AB6221)

The network-switch for the PROFINET 2-port bus structure is integrated in the bus module. With this solution the connection to the field bus is done via the flange plate with the two integrated M12 female plugs (D-coded).

LED function PROFINET 2-Port (AB6221):
Front view:

No.	Description
1	Network Status LED
2	Module Status LED
3	Link/Activity LED (port 1)
4	Link/Activity LED (port 2)



Network Status LED

LED State	Description	Comments
Off	Offline	• No power • No connection with IO controller
Green	Online (RUN)	• Connection with IO controller Established • IO Controller in RUN state
Green, flashing	Online (STOP)	• Connection with IO controller EstablishedIO Controller in STOP state

Module Status LED

LED State	Description	Comments
Off	Not initialized	No power - or - Module in 'SETUP' or 'NW_INIT' state
Green	Normal operation	Module has shifted from the 'NW_INIT' state
Green, 1 flash	Diagnostic event(s)	Diagnostic event(s) present
Green, 1 Hz	Blink	Used by engineering tools to identify the node on the network
Red	Exception error	Module in state 'EXCEPTION'
Red, 1 flash	Configuration error	Expected Identification differs from Real Identification

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LED State	Description	Comments
Red, 2 flashes	IP address error	IP address not set
Red, 3 flashes	Station name error	Station Name not set
Red, 4 flashes	Internal error	Module has encountered a major internal error

LINK/Activity LED

LED State	Description	Comments
Off	No Link	No link, no communication present
Green	Link	EtherNet link established, no communication present
Green, flickering	Activity	EtherNet link established, communication present

8.3.1.4 PROFIBUS (AB6200)

For the PROFIBUS structure the connection to the field bus is done via the flange plate with the two integrated M12 female/male plugs (B-coded). The bus termination on both ends for the PROFIBUS can be turned on and off with the switch in the 9-pole D-Sub connector. External termination resistors are not possible, because the 5 V voltage is missing in the M12 plugs.

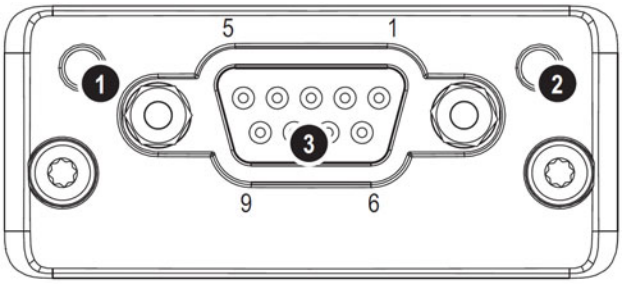


With the activation of the bus termination the outgoing cable is disconnected at the plug!

The male plug in the flange plate is for the incoming cable and the female plug for the outgoing cable.
A valid address range from 0 to 125 is possible. The address 126 is reserved for the SSA (Set Slave Address) function.

LED function PROFIBUS (AB6200):
Front view:

No.	Description
1	Operation mode
2	Status
3	PROFIBUS connector



Operation Mode

State	Indication	Comments
Off	Not online / no power	
Green	Data exchange	
Green, flashing	Clear	
Red flashing (1 flash)	Parametrization error	

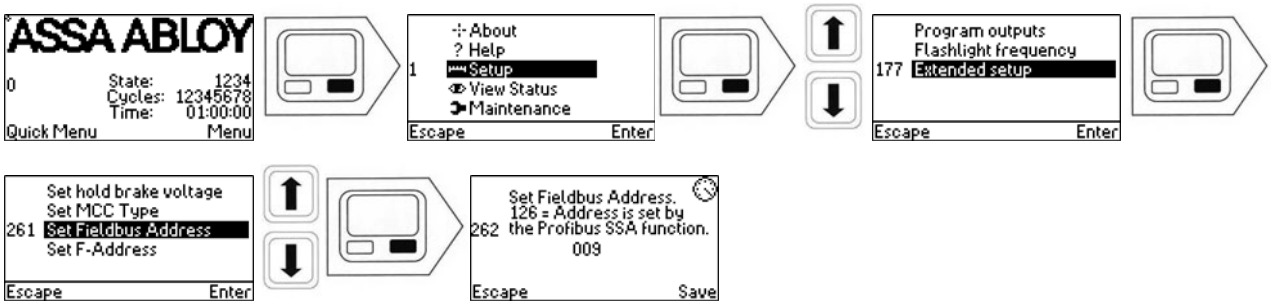
State	Indication	Comments
Red flashing (2 flashes)	PROFIBUS configuration error	

Status

LED State	Description	Comments
Off	Not initialized	Anybus state = "SETUP" or "NW_INIT"
Green	Initialized	Anybus module has left the 'NW_INIT' state
Green flashing	Initialized, diagnostic event(s) present	Extended diagnostic bit set
Red	Exception error	Anybus Module in state 'EXCEPTION'

8.3.2 Set Fieldbus Address [1]

The address change will take first effect after the control system is switched off and on again.



8.3.3 Generic Station Description (GSD-file)

The Generic Station Description (GSD-file) of the used bus module can be downloaded at http://www.anybus.com. Search for the appropriated type.
Observe the user manual and the specifications of the manufacturer!

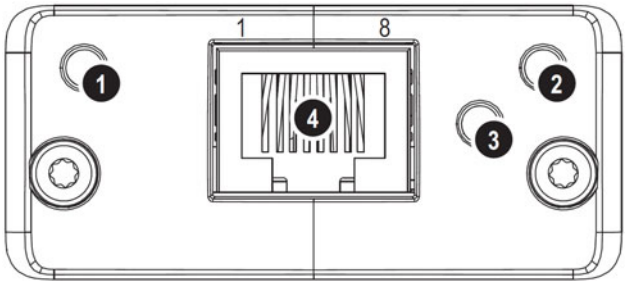
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8.3.4 EtherNet/IP (AB6214)

The TCP/IP configuration setup can be made with the Anybus IP config Utility for module TCP/IP configuration. The utility as well as the EDS file can be downloaded from http://www.anybus.com.

LED function EtherNet/IP (AB6214):
Front view:

No.	Description
1	Network Status LED
2	Module Status LED
3	Link/Activity
4	EtherNet Interface



Network Status LED

A test sequence is performed on this LED during startup.

LED State	Description
Off	No power or no IP address
Green	On-line, one or more connections established (CIP Class 1 or 3)
Green, flashing	On-line, no connections established
Red	Duplicate IP address, FATAL error
Red, flashing	One or more connections timed out (CIP Class 1 or 3)

Module Status LED

A test sequence is performed on this LED during startup.

LED State	Description
Off	No power
Green	Controlled by a scanner in Run state
Green flashing	Not configured, or scanner in Idle state
Red	Majar fault (EXCEPTION-state, FATAL error etc.)
Red flashing	Recoverable fault(s)

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LINK/Activity LED

LED State	Description
Off	No link, no activity
Green	Link established
Green, flickering	Activity

8.3.4.1 Setup of the EtherNet/IP-Master

The settings shall be done according to the table below, which conforms to the information in the EDS file. These terms are also used in the RSLogix 5000.

	Assembly Instance	Size	Format
Input	100	2	16 bit
Output	150	2	16 bit
Configuration	5	0	8 bit

8.3.5 Bus communication

The cyclic write and read process data, in PROFIBUS and PROFINET called IO data will be mapped in to so call ADI's, application data instances. These are 16bit unsigned variables that can also be used for acyclic data when mapped so.

8.3.6 TX data from the MCC 2

8.3.6.1 Data format

IO data (as seen by the PLC master)	Size	Format
Input (ADI 0)	1	16 bit (Word)
Input (ADI 1)	1	16 bit (Word)
Output (ADI 2)	1	16 bit (Word)
Output (ADI 3)	1	16 bit (Word)

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8.3.6.2 Application Data Instance ADI[0]

Byte	**2**								**1**							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Bit position	**Rules**	**Value**	**Comment**
1: Door ready for work	0: Not ready 1: Ready	0x0001	
2: Door is closed	0: Door is not closed 1: Door is closed	0x0002	
3: Door is open	0: Door is not open 1: Door is open	0x0004	
4	0: Door is not open to reduced height 1: Door is open to reduced height	0x0008	
5	For future use	0x0010	
6	For future use	0x0020	
7	For future use	0x0040	
8	For future use	0x0080	
9: Custom 1		0x0100	Custom1-8: For customized digital data out from the MCC 2 (see electrical drawing).
10: Custom 2		0x0200	
11: Custom 3		0x0400	
12: Custom 4		0x0800	
13: Custom 5		0x1000	
14: Custom 6		0x2000	
15: Custom 7		0x4000	
16: Custom 8		0x8000	

8.3.6.3 Application Data Instance ADI[1]

Byte	**2**								**1**							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

The ADI[1] data contains a 16 bit unsigned value of the current state of the MCC 2 door controller. These data can be used for detecting normal operation states and for the error messages. A list of all the error states follows below.

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8.3.6.4 Allocation of the MCC 2 state messages

MCC 2 state messages	Decimal state number	Hexadecimal state number
Only activated shortly during boot	0	0
Initiating after boot/ Power on.	1	1
Checking if any up, stop or close commands is active during power up. Remains in this state if so.	2	2
Select if reference search is necessary or not. If the display stop function was activated before reboot will the control stay in this state until the display up or down button is activated.	3	3
Waiting for a display unit open button activation to run to the reference position	5	5
Waiting for any open or close signal to start reference search	6	6
Waiting in open position for the hold open time to count down	7	7
Waiting in open position for prerunning time (traffic light/ warning lights on) to count down	8	8
Testing the safety functions before closing.	9	9
Closing a small distance.	10	A
Reversing direction due to activated safety functions or open signals	11	B
Waiting for a display unit close button activation to run to the reference position	12	C
Closing a small distance in reference search mode.	13	D
Opening after reversing due to activated safety functions or open signal	14	E
Opening to the reference limit	15	F
The state is not used yet.	16	10
The door needs to be closed in hold-to-run mode with the display unit.	16	10
Opening to the reference limit after an open signal without closing after time	17	11
Opening to the reference limit to get a valid position after a close signal, will be followed immediately by closing in normal mode to the closed position.	171	AB
The door is stopped after an error that has been reset.	20	14
The door is closed	30	1E
Testing the opening height by spring power. Only used for the RR300 plus F+R door.	31	1F
Opening with motor power after testing the opening height by spring power. Only used for the RR300 plus F+R door.	32	20

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MCC 2 state messages	Decimal state number	Hexadecimal state number
Preruning time before opening in close after time mode with warning/ traffic lights on	41	29
Preruning time before opening to reduced height in close after time mode with warning/ traffic lights on	43	2B
Opening after the close after passage function has been activated	49	31
Opening in close after time mode after the close after passage function has been activated	491	1EB
Opening to reduced height after the close after passage function has been activated	492	1EC
Opening to reduced height in close after time mode after the close after passage function has been activated	493	1ED
Opening after an open signal	50	32
Opening after an open signal with closing after time	51	33
Opening after an open signal to reduced height	52	34
Opening after an open signal to reduced height with closing after time	53	35
Reversing direction due to activated safety edge or prerunning photocell	54	36
Reversing direction in closing after time mode due to activated safety edge or prerunning photocell	541	21D
Reversing direction in closing after time mode due to a broken safety edge or prerunning photocell	542	21E
Reversing direction due to an open signal or stationary photocell activation	55	37
Reversing direction in closing after time mode due to an open signal or stationary photocell activation	551	227
Reversing direction due to an open signal to reduced height.	552	228
Reversing direction due to an open signal with closing after time mode to reduced height.	553	229
Reversing direction after a permanent open signal from the display unit or an external button	554	22A
Reversing direction after a permanent open signal from an external switch	555	22B
Reversing direction in close after passage mode after a stationary photo cell activation	556	22C
Reversing direction in close after passage mode with closing after time after a stationary photo cell activation	557	22D
This state is not used.	559	22F
The door is open and will close after a close signal.	56	38

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MCC 2 state messages	Decimal state number	Hexadecimal state number
The door is open and will close after the hold open time if no open signals or safety functions are keeping it open.	57	39
The door is open at reduced height and will close after a close signal.	58	3A
The door is open at reduced height and will close after the hold open time if no open signals or safety functions are keeping it open.	59	3B
The door is open and will close after the close after passage time has passed if no safety functions are keeping it open.	60	3C
The door is open and will close after the close after passage time or the hold open time if no safety functions are keeping it open.	610	262
The door is open to reduced height and will close after the close after passage time has passed if no safety functions are keeping it open.	611	263
The door is open to reduced height and will close after the close after passage time or the hold open time if no safety functions are keeping it open.	612	264
The door is opening after a permanent open signal from the display unit or an external button	61	3D
The door is opening after a permanent open signal from an external switch	62	3E
The door is open after a permanent open signal from the display unit or an external button	63	3F
The door is open after a permanent open signal from an external switch	64	40
Prerunning time before closing with the traffic/ warning lights on	70	46
Check and test of the safety functions before closing.	73	49
Check and test of the safety functions before closing in closing after hold open time mode	74	4A
The stationary photocell or the prerunning photocell or an additional safety function was active more than two seconds after a close input signal was given and the door could not close.	75	4B
The door is closing.	80	50
The door is closing in closing after time mode.	81	51
A stop input function has stopped the door.	100	64
The display unit stop button has stopped the door. Can only be reset by pressing the open or close button on the display unit.	101	65
The emergency stop input (X72 or X73) has stopped the door.	102	66

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MCC 2 state messages	Decimal state number	Hexadecimal state number
The door is hindered to move by the interlock or airlock function, meaning that the other door in interlock or airlock is open.	150	96
The door is in hold to run mode. The door only runs in hold to run through the display buttons.	200	C8
The door is open in hold to run mode. The door only runs in hold to run through the display buttons.	210	D2
This state is not used.	220	DC
The door is in position setup mode and the door closed position shall be set. The door only runs in hold to run through the display buttons.	500	1F4
The door is in position setup mode and the door open position shall be set. The door only runs in hold to run through the display buttons.	501	1F5
The door is in position setup mode and the hold open time shall be set. The door only runs in hold to run through the display buttons.	502	1F6
The door is in position setup mode and the breakaway reset position shall be set. Used for doors with a special position for breakaway reset.	503	1F7
The door is in position setup mode and the door closed position shall be set. The door only runs in hold to run through the display buttons. This state is used for doors which have the position reference in the door closed position, like pneumatic safety edge.	510	1FE
The door is in position setup mode and the door open position shall be set. The door only runs in hold to run through the display buttons. This state is used for doors which have the position reference in the door closed position, like pneumatic safety edge.	511	1FF

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8.3.6.5 Allocation of the MCC 2 error messages

MCC 2 error messages	Decimal error state number	Hexadecimal error state number
Electrical safety edge / pre-running safety photocell interrupted	1004	3EC
Electrical safety edge or the pre-running safety photocell activated	1005	3ED
Thermal contact tripped	1006	3EE
Encoder gives incorrect signals	1009	3F1
Encoder registers unauthorised movement	1010	3F2
Emergency lever/hand crank actuated	1011	3F3
Broken wire/circuit to Reference limit switch	1012	3F4
Reference limit switch circuit interrupted	1013	3F5
Crash sensor has tripped	1015	3F7
Safety 1 defect	1018	3FA
Door line photocell interrupted	1019	3FB
Safety 2 defect	1022	3FE
Internal frequency converter error	1023	3FF
Safety edge interrupted	1024	400
Internal error	1025	401
Reference limit switch interrupted	1026	402
Error in safety edge input	1027	403
Error in emergency stop input	1028	404
Input voltage at emergency stop input too high	1029	405
Earth fault in the 24 V circuit (low voltage)	1030	406
Earth fault in the 24 V circuit (high voltage)	1031	407
Opening height not reached during cyclical self-test (only for RR300 Plus F+R)	1032	408
Breakaway function activated	1033	409
Frequency converter undervoltage	1100	44C
Frequency converter phase error	1101	44D
Frequency converter overvoltage	1102	44E
Frequency converter overvoltage warning	1103	44F
Frequency converter error, wrong main voltage	1104	450
Frequency converter overcurrent	1105	451
Frequency converter error, motor blocked	1106	452
Frequency converter error, IGBT overload	1108	454
Frequency converter error, wrong hardware	1110	456
Frequency converter error, brake circuit overcurrent	1112	458
3 phase error, frequency converter	1115	45B
Electrical safety edge or pre-running safety photocell activated	1200	4B0

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8.3.7 RX data, data to the MCC 2

8.3.7.1 Application Data Instance ADI[2]

Byte	**2**								**1**							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1



For a RESET the control system needs to be restarted include door reference run.

Bit position	**Rules**	**Value**	**Comment**	**Reset**
1: Open door (UP)	0: NOP 1: Open the door	0x0001		Lost bus communication or the Anybus module is in faulty
2 :Stop door (Stop)	0: NOP 1: Stop the door	0x0002		Lost bus communication or the Anybus module is in faulty
3: Close the door (close)	0: NOP 1: Close the door	0x0004		Lost bus communication or the Anybus module is in faulty
4: Open the door to reduced height	0: NOP 1: Open the door to reduced height	0x0008		Lost bus communication or the Anybus module is in faulty
5 Reset system	0: NOP 1: The system will be reset Condition: "Door ready for work" message = 0	0x0010	This bit is used to reset the control from error states. This bit shall only be active for a maximum time of 500ms to avoid that the control is restarted several times.	Lost bus communication or the Anybus module is in faulty
6	For future use	0x0020		
7	For future use	0x0040		
8	For future use	0x0080		

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Bit position	Rules	Value	Comment	Reset
9: Custom 1		0x0100	Custom1-8: For customized digital data in to the MCC 2 (see electrical drawing)	Custom1-8: Lost bus communication or the Anybus module is in faulty
10: Custom 2		0x0200		
11: Custom 3		0x0400		
12: Custom 4		0x0800		
13: Custom 5		0x1000		
14: Custom 6		0x2000		
15: Custom 7		0x4000		
16: Custom 8		0x8000		

8.3.7.2 Application Data Instance ADI[3]

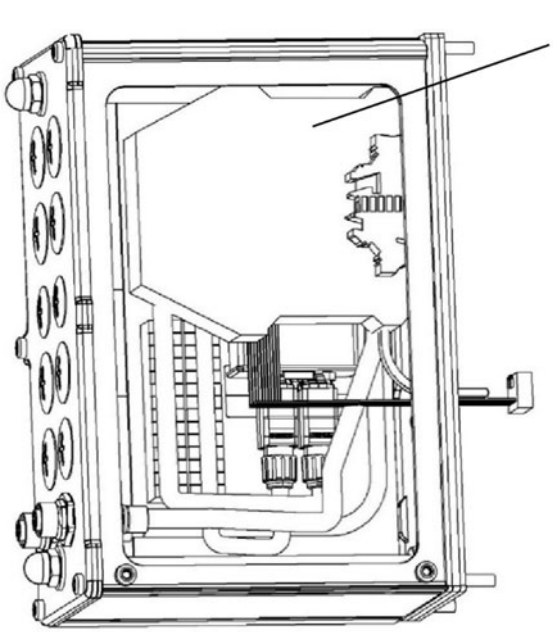
Byte	**2**								**1**							
Bit	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Bit Pos.	Rules	Value	Reset
1	For future use	0x0001	Lost bus communication or the Anybus module is in faulty
2	For future use	0x0002	
3	For future use	0x0004	
4	For future use	0x0008	
5	For future use	0x0010	
6	For future use	0x0020	
7	For future use	0x0040	
8	For future use	0x0080	
9	For future use	0x0100	
10	For future use	0x0200	
11	For future use	0x0400	
12	For future use	0x0800	
13	For future use	0x1000	
14	For future use	0x2000	
15	For future use	0x4000	
16	For future use	0x8000	

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8.3.8 Installation of the safety relays (Option for machine protection doors)

Additional to the bus module it is possible to mount a safety relay (1) for the safety limit switches at the machine protection door in the MCC 2 extension box. This relay is snapped on the DIN rail of the MCC 2 adapter plate. The wiring is done via the terminal blocks X308a and b on the mainboard.



Extension with bus module and safety relay

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8.4 PFC filter

Danger: Risk of electrical shock!

Dangerous DC voltage (220-850V DC) can apply on the PFC circuit, which can lead to an electric shock with serious injuries or death.
Install or work only on the PFC filter if the MCC 2 is turned off for min. 30 seconds.
Use only wires with an insulation voltage of 1000 V DC.
The wire of the PFC filter must meet at least U0/U 0.6/1 kV according to DIN VDE 0298 and IEC 183 and a min. cross section of 1,5 mm^2.

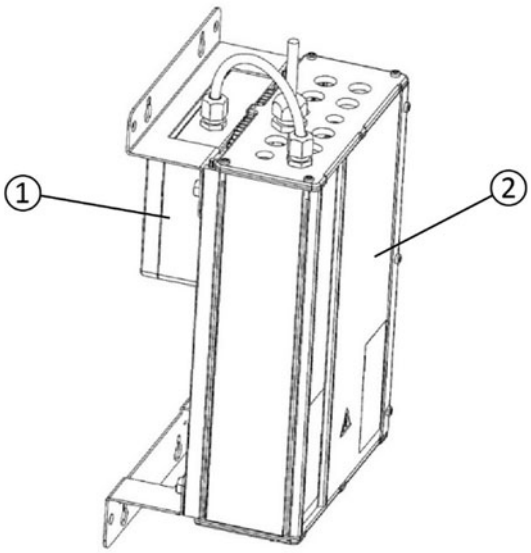
Notice: Loss of the CE conformity.

Without PFC filter the requirements of the European EMC directive for residential, commercial and light industry environments are not kept and it may come to disturbances of other electrical devices.
PFC filter is required for residential, commercial and light industry environments.

For application without PFC filter there needs to be a bridge on X28 between pole 1 and 2. This should have a min. cross section of 1,5 mm^2.

In order to keep the EMC requirements for the residential, commercial and light industry according to DIN EN 61000-6-3 an additional PFC filter is necessary.
The PFC filter is installed in a separate housing and mounted below the MCC 2. Electrical wiring is to be made between MCC 2 and the filter. Observe the electrical drawing of the door for details.

1) PFC filter
2) MCC 2



MCC 2 with PFC filter

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